

DUAL ABSOLUTE LEAK TESTER



T8090 USER MANUAL



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Certificate of Conformity CE

Manufacturer: ForTest via SibillaAleramo 7 (sez. 2A) - 41123, Modena, Italia

Model: T8090

Description of the equipment: leak tester instrument.

Date of issue: 27/11/2019

ForTest declares that the instrument T8090, object of this manual, complies the following standards:

-) 2014/35/UE Low voltage directive

Applied norms:

- EN 61010-1

-) 2014/30/UE Electromagnetic compatibility (EMC) directive

Applied norms:

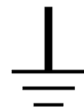
- EN61326-1/ EN61326/A1
- EN61000-3-2/ EN61000-3-2/A14
- EN61000-3-3/ EN61000-3-3/A1
- EN61000-4-2/ EN61000-4-2/A1
- EN61000-4-3/ EN61000-4-3/A1/ EN61000-4-3/A2
- EN61000-4-4
- EN61000-4-11

Then ForTest guarantees the use of the equipment object of the present manual with its higher safety. Moreover, it is guaranteed that the design of the equipment object of the present manual has been made to protect the operator from:

- Electrical shock
- Mechanical hazards
- Flame propagation beginning from the equipment
- Mechanical resistance to heat
- Gas emissions, explosions and implosions in the following ambient conditions:
 - Internal use
 - Ambient temperature of working from 5°C to 40°C ($\pm 2^\circ\text{C}$)
 - Max relative humidity of 92,5% ($\pm 2,5\%$)

Warranty is not valid in case you do not respect the prescribed conditions or in case of a usage against the rules of the interested job.

T8990 leak tester (only in 24 VDC version) is equipped by a terminal of protection, placed on the back of equipment and connected inside in compliance with safety standard. This terminal is necessary to connect from an external system to protection connector. Terminal is recognizable from the symbol:



ForTest is not responsible in case of noncompliance of the conditions above mentioned.

Chief executive
Alessandro TOMASINI

General Warranty Conditions

- Performances under guarantee must be made exhaustively from ForTest authorized staff.
- Excluded the clauses written in this warranty certificate none is authorized to give any other, written or oral ones.

1) Duration

ForTest guarantees the user:

- 36 months warranty on T8090 for all the components or parts that can present manufacturing defects.
- On the modified equipment of the T series the limits prescribed by law are used.

2) Effect

- The warranty period begins from the date of production of the equipment, verifiable by the date written on the transport document.

3) Warranty development

- The possible intervention of first start and configuration of the equipment is totally charged to the Customer.
- Warranty implies the reset or the substitution of the faulty parts and the respective manpower.
- During the warranty period, the Customer has to pay to the authorized staff that makes the intervention the relative charges concerning the possible transfer.
- Interventions, under warranty or out of warranty, made on the equipment must to be made by ForTest authorized staff. Otherwise the terms expected on the equipment fall down immediately.
- Warranty does not cover damages to the components of the equipment caused by:
 - Bad maintenance.
 - Bad or wrong setting.
 - Bad or improper quality of the compressed air put into the equipment.
 - Possible electrical or pneumatic system not in accordance with laws in force.
 - Carelessness or misuse.

- Electrical power with instantaneous voltage values or continuous one out of rule.
- Impacts or falling down of foreign bodies.
- It should be noted that the opening of the box of the equipment, together with the conditions above described, will cause the immediate revocation of the guarantee.
- Removal of the safety adhesive seal will immediately invalidate the warranty.
- Passed or fall warranty conditions, the user must pay not only the possible transfer of the authorized staff, but also the manpower and the cost of the replaced parts.
- Authorized staff will supply to the Customer, on his special request, the possible replaced pieces with in attached the reasons of the repairing.

Charges due to the manpower will be paid by the Customer whenever the requested intervention has been caused by bad maintenance or incorrect installation of the equipment itself.

Safety Considerations

Read carefully before using or connecting the instrument



WARNINGS: The use of gases at pressures other than the ambient one involves risks deriving from the force that the gas can assume. Whatever the application, test parts, piping, capping, represent a possible source of risk caused by danger to humans, things and the surrounding environment. It is therefore recommended to always check the pneumatic circuit, to put it in protection with the help of barriers or any system for the purpose. It is recommended to instruct personnel on the risks of using pressure gas, as well as on the use of instrumentation and how to test parts. ForTest Italia s.r.l. guarantees the safety of its products (if used with the gases and pressures for which they were built, and if conducted by qualified personnel), but declines all responsibility for incorrect or improper use of the equipment or failure to comply with the basic instructions contained in this manual.

Always follow the basic precautions below to avoid the risk of serious injury or even death resulting from electric shock, short circuit, damage, fire, or other hazards. These precautions include, but are not limited to, the following:

1) Power supply / power supply cable

- Use only the correct voltage specified for the instrument. The required voltage is indicated on the label on the instrument.
- Check the plug periodically and remove any dirt or dust that may have accumulated.
- - Use only the power supply cable, power supply, battery, charger supplied by ForTest Italia s.r.l.
- If present, do not disassemble or handle the Lithium battery pack.
- Do not place the power supply cable near any heat sources, where it may be walked on or damaged. Do not place heavy objects on the power supply cable.
- Connect the cable to an appropriate grounded outlet. Improper grounding may result in electric shock.
- If you do not intend to use the instrument for a long period of time or in case of thunderstorms, remove the plug from the electrical outlet.

2) Operator

- ForTest Italia s.r.l., recommends that the operator intended to use the appliance has adequate training and a level of qualification for the work required and has read the instructions for use, safety and maintenance indicated in this manual.

3) Positioning

- Do not place and use the instrument in an unstable position to prevent it from falling.
- Do not place and use the instrument near any heat sources.
- Do not expose the instrument to rain, use near water or in conditions of high humidity, do not place containers with liquids on the instrument. Leaking liquid into the openings of the instrument can cause damage, malfunction and risk of electric shock to humans. In the event of liquid leaking into the instrument, such as water, immediately disconnect the instrument from the power supply.

4) Pneumatic connection

- Do not feed the instrument with pressures higher than those indicated in the "Back panel" chapter of this manual.
- Make the pneumatic connections as indicated in the chapter "Pneumatic connections" in this manual.
- Use properly filtered and dried air, follow the instructions on air quality given in the chapter "Back panel" of this manual.

5) General considerations

- Do not use this instrument until you have read and observed the points listed above in this chapter.
- Please read this manual carefully before use.
- All electrical connections to the device must be equipped with safety systems that meet the requirements and comply with the applicable standards.
- To avoid electromagnetic interference, the electrical connections made on the device must be less than 2 m in length.
- If present, the electrical power socket must be equipped with an earth connection that complies with the regulations in force.

- Disconnect the power supply (electrical outlet, connection cable to the power supply, battery charger, battery) before performing any maintenance work.
- Switch off the compressed air supply during any work on the tyre mounting.
- Do not open the device, otherwise the warranty will be void.
- Avoid direct contact of the appliance with water.
- -Do not use accessories (other than those supplied) of other brands without the express permission of ForTest Italia s.r.l.
- Do not remove the identification label on the instrument.

ForTest Italia s.r.l. is in no way responsible for accidents caused by failure to comply with safety rules and incorrect use of the measuring instrument.

Introduction

Through the revolutionary Dual Absolute® technology, ForTest combines the simplicity of a classic absolute drop system with the performance of a differential system, taking leak test instrument technology to a level never before achieved.

Equipped with multiple operating modes and the hardware and pneumatic optimization T8090 allows:

- Eliminate thermal and mechanical drift phenomena in the Reference part, to obtain the most stable, true and repetitive measurement
- Testing two pieces at a time, halving cycle times
- Completely eliminate the periodic maintenance required in classic differential instruments
- Ensure double safety of the measuring channels, based on absolute drop circuits with intrinsic safety and cross-checking between channels
- Achieve full scale measurement up to 200 bar, contrary to traditional limits of differential systems
- Reduce global testing time

Mode of function T8090

Absolute TEST

This mode of operation makes it possible to perform the leak test by detecting the pressure drop in a programmed test time.

Basically, the test piece is filled at a certain filling pressure, left to settle to stabilize the pressure inside the piece and then carried out the pressure variation measurement in a certain time, called test time.

Only one transducer on the test port side is in operation during this test mode.

Differential

This mode of operation allows the leak test to be performed by detecting the pressure difference between a reference master piece and the piece to be tested.

In this case, during the whole filling phase, both pieces are filled, after which we pass to the settling phase where for a programmable time in the parameter "**Settling Equalization**" a percentage is set on this phase (initial phase) in which the Reference piece and the test piece are in communication, thus performing a better balancing of the pressures in the two pieces before entering the test phase, where the pressure difference between the two pieces is detected.

Subsequently a settling phase is carried out to allow the air inside the two pieces to settle after filling. In this phase it is possible to program (through the parameter "**Equalization in settling**") a time within which the two reference and test pieces remain in communication, thus performing a further pressure balancing.

The final part of the cycle ends with the test time where the leak is detected.

Generally this mode allows to intercept lower leakage rates than the absolute drop mode, mainly on rigid pieces with little expansion.

Dual Absolute



This mode of operation allows the leak test to be performed by detecting the pressure difference between a reference master piece and the piece to be tested. Unlike the purely differential mode, however, the reference piece is filled periodically, setting a specific frequency range depending on its conformation.

The pressure trend in the reference piece is then sampled at regular intervals and the final result (difference between the loss trend of the reference piece and the piece under test) is calculated starting from the sampled values and no longer in real time on the reference piece itself.

In this way it is possible to guarantee a behaviour of the reference part much more similar to the test part which is actually filled and tested only once.

By programming an adequate sampling frequency of the reference part, it is possible to leave it to return to the initial conditions, in terms of expansion and pressure variations, as if it were tested for the first time, exactly like the part to be tested.

The difference in pressure between the test piece and the reference piece is then calculated on the last sampling detected on the reference piece.

This mode is therefore suitable for elastic parts with medium to large volumes.

Zero Center **X2**

This mode of operation allows the leak test to be carried out on two pieces at the same time, considering the difference in pressure between the two pieces as discriminating.

The instrument uses advanced algorithms to avoid giving "good" results in case of similar pertitis on both pieces under test. This is also thanks to the DUAL ABSOLUTE system, i.e. exploiting the principle of absolute drop measurement on both TEST channels (TEST PORT and REFERENCE).

Through this mode of operation it is possible to almost halve the test cycle and therefore production times compared to traditional testing systems.

The T8090 series is ForTest's standard range of leak testing and flow testing instruments.

A pneumatic circuit optimized to cancel the parasitic test volume, an electronic board that uses the latest technologies, the latest in microprocessor and integrated circuits and a firmware expandable but at the same time optimized allow you to perform reliable tests and never so fast.

The panel is completed by the graphic display of 800x280 pixels in colour, 8 control keys, the front panel in tempered glass and the pressure regulator on the front, all enclosed in an elegant and resistant case, with an anodized aluminium back panel that integrates pneumatic elements and electrical connectors. The display shows a test graph that allows you to see the trend of the quantities measured by the instrument during the test.

The T8090 series is equipped as standard with a 15-pin connector for 24 Vdc input signals and medium power outputs, a USB port on the front panel, two optoisolated RS232 ports, and a port for optoisolated RS 485 multipoint industrial networks. External accessories such as printers, network sorter kits for complex PC networks, electronic pressure regulators, barcode readers, Bluetooth connection, Ethernet connection, WiFi connection, etc. are available.

The T8090 series, thanks to a renewed hardware, boasts a measurement resolution of 0.1 Pa that allows you to perform tests with greater speed and reliability. The simple user interface, the new touch screen keyboard, the multi-language menu, the bargraph, the RGB LEDs for indication of good or rejected part, complete the rich equipment.

Frontal Panel

The frontal panel is constituted by different elements (eventually also optional) of interface with the operator. Here below we make a short description, but for more information it is better to read carefully the chapter *Test Menu* and *Setup Menu*.



STAUBLI® JUNCTION (5)

Staubli® junction for connection to ForTest M0710 calibrated leak or of other constructors.

PRESSURE REGULATOR (2)

If it is present, it allows to set filling pressure of the piece under test. According to the full scale of the equipment, different models can be mounted. In case of an electronic pressure regulator is mounting (optional) it will not be present on the panel any regulation knob.

LED (4)

The double bar of blue led visualizes, typically, upward the pressure trend and on the bottom bar the secondary measure trend of the leak.

If the equipment is with the flashing result at display, also led bars will be flashing (with a brightness modulated in time, in order to disturb the eyes as little as possible), otherwise they remain fixed at its maximum brightness.

In case that a transducer has reached its full scale, the bar relative to the transducer in object will have $\frac{3}{4}$ of the led flashing.

At the right of the double bar of led there are the outcome indicator lights: red for the waste and green for the good.

At the end of a test, its corresponding result will make flashing the led colour combined with the result: the whole with an intensity modulated in time, in order to disturb the eyes as little as possible.

During a test, in case the secondary measure is referred to a time (pressure/time or volume/time), the result led will give a partial runtime result, but a low brightness.

TOUCH SCREEN KEYBOARD (3)

Here below we describe the meanings of the frontal buttons and of its use.

Button [TST]

It allows to enter or to exit the setting menu of testing parameters (see chapter Test Menu).

If you are outside the Test Menu, the pressure of this button allows to enter into the menu, of which you will see at display the first screen; if otherwise we are already inside the menu, the pressure of the button will cause the exit from the menu.

Button [SET]

It allows to enter or to exit the parameters setting menu of the equipment (Setup Menu chapter) and at the menu of summary of the specific parameters of the equipment (chapter Version Menu).

If you are outside both of the Setup Menu and of the Version Menu, the pressure of this button allows to enter the Setup Menu of which you will see the first screen at display. To enter the Version Menu, it is necessary to press the button [SET] twice. If we are already inside one of the two menus (excluded the previous case that allows to enter the Version Menu), the pressure of the button will cause the exit from the menu in which you are.

Button [CTP]

It allows to enter or to exit from Counting Piece Menu (Counting Piece Menu chapter).

If we are outside the Counting Piece Menu, the pressure of this button allows to enter the menu, of which you will see at display the first screen; if otherwise we are already inside the menu, the pressure of the button will cause the exit from itself.

Button [▲]

If we are in a menu on the navigation arrows of a submenu up on the right, the pressure of this button allows to scroll behind the previous screen of the menu, that is to visualize the previous sub menu (if we are not already on the first sub menu).

If we are in a menu and in particular on a parameter (the parameter value will flash), the pressure of this button allows to scroll behind the previous parameter, without confirming the value possibly modified.

If the equipment has finished a test and it is showing the result at display, this button allows to scroll all the significant measures stored by the equipment as characteristics of the test (obviously only if they are more than two ones).

In case of electronic regulator, this button has also some other functions, we send you to the chapter relative to the optional for a description of them.

Button [▼]

If you are in a menu on the navigation arrows of a sub menu up on the right, the pressure of this button allows to scroll towards the following screen of the menu, that is to visualize the following submenu (if you are not already on the last sub menu).

If you are in a menu and particularly on a parameter (the parameter value will flash) the pressure of this button allows to scroll towards the following parameter, without confirming the value eventually modified.

If the equipment has finished a test and it is showing its result, this button will allow to scroll all the significant measures stored by the equipment as characteristics of the test (obviously only if they are more than two ones).

In case of electronic regulator, this button has also some other functions, we send you to the relative chapter of the optional for their description.

Button [+]

This button has several functions according to the state in which the equipment is.

If you are not in any menu, and if there are not any tests running, the pressure of this button allows to give the start to the test, if in the Setup Menu the relative parameter has been set properly.

If you are in a menu and in particular on a parameter (the parameter value will flash), the pressure of this button allows to increase the value (if the maximal value has not been reached yet).

Button [-]

This button has several functions according to the state in which the equipment is.

If you are not in any menu, and there are not any tests running, the pressure of this button allows to make the autozero.

If you are not in any menu, and there is a test running, then the pressure of this button allows to give abort to the test (whatever it is the way to give the start).

If you are in a menu and particularly on a parameter (the parameter's value will flash), the pressure of this button allows to decrease the value (if its minimum value has not been reached yet).

AUTOZERO

Autozero is a proceeding by which the equipment keeps as a reference of zero of measure that one that it is measuring actually (attention, without replacing the calibrating value stored in the equipment, but only setting a kind of temporary calibration).

This function is very powerful (it allows to have a continuous calibration of the equipment) but also to use with knowledge (the equipment firmware has some protections against any abuses of this function). We advise, therefore, to use this function when you are sure that there is not any residual pressure in the piece under test, or (advised) there is anything connected to the TEST PORT.

Autozero proceeding, can be started by the button or automatically by the equipment: see the chapter Setup Menu.

Starting an auto zero proceeding, make sense if the equipment has a discharge solenoid (outside or inside) or a solenoid dedicated to auto zero: otherwise, the pneumatic circuit cannot physically empty the piece or connect it to the ambient pressure in order to keep its reference.

Button [E]

This button has several functions according to the state in which the equipment is.

If you are on a parameter, the pressure of this button allows both to confirm the value, and to pass to the modification of the following parameter: in case you are on the last parameter of the actual screen, after the confirmation of the actual value, you will go to the navigation arrows of the submenu up on the right.

If you are on the navigation arrows of the sub menu up on the right, the pressure of this button allows to enter into the sub menu visualized, that is to pass at the modification of the first parameter of the visualized submenu.

If the equipment is in flashing result state at display, the pressure of this button allows to come back to the waiting screen.

USB KEYPAD (6)

Allows connection to the storage medium (USB key).

MECHANICAL BUTTON (7)

Allows you to start the test, if the instrument is waiting to start and the "Start" parameter of the setup menu is set to "+", or to abort the test if it is in progress.

DISPLAY (1)

According to the state of the equipment, the display shows the information with different layout.

Instrument in a menu:

TEST MENU: PHASES TIME, PRG:	001/1	▲	▼
Filling phase duration:	5.0	[s]	
Settling phase duration:	5.0	[s]	
Test phase duration:	5.0	[s]	
Discharge phase duration:	1.0	[s]	
Program name: TEST			

On the top left you see the title of the submenu, on the right the indicator of the navigation arrows to scroll through the submenus that make up the menu, and you can then by pressing the keys [▲] or [▼], scroll to the previous or next submenu, or by pressing the key [E], access the modification of the parameters that make up the current submenu.

In the example, that refers to a screen of Test Menu, you can see next to the navigation arrows the program number activated (this only for this menu, see the chapter *Test Menu*).

Below line instead can has on the left data and hour if you are in Setup menu or Countpieces, or the name of the used program if you are in Test menu, while on the right always the 4 icons that indicate:



Indicates that the wireless card is turned off.



Indicates that the Bluetooth 2.0 signal is active.



Indicates that the Bluetooth Low Energy (BLE) signal is active.



Indicates that a storage medium (e.g. USB key) is connected, at the end of each test this icon flashes to indicate that the saving has taken place. If the storage medium is not connected, the icon will be crossed out.



Indicates that the instrument is not locked by a password and that therefore it is possible to modify the parameters of TST and SET and change the program.



Indicates that the instrument is locked by password, the parameters of the test and setup menus in this condition are not modifiable by the user.



Indicates that the instrument is locked by password, the parameters of the test and setup menus in this condition are not modifiable by the user. Furthermore, it is not possible to change the test program.



Indicates that the instrument is locked by password, the parameters of the test and setup menus in this condition are not modifiable by the user. Moreover, it is not possible to change the test program and it is not possible to access the test menu.



Indicates that there are no system errors.



Indicates Dual Absolute mode is active



Indicates that patency mode is active



Indicates that Differential mode is active



Indicates that Absolute Test mode is active.



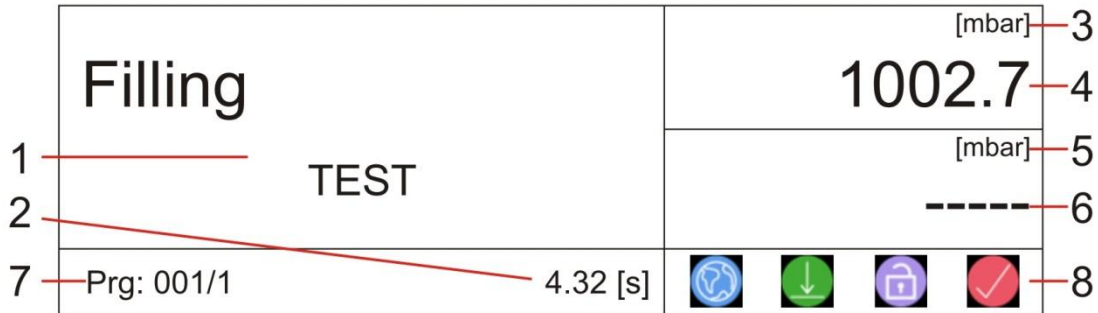
Indicates that Zero-Center mode is active

When you are on a parameter, it will be highlighted and it is possible to modify it with **[+]** and **[-]** buttons or, only in case of numeric value, through touch screen keyboard that has this layout:



At this point, to set up a numeric value for a particular parameter, it is enough to set the wanted value and press **[E]** button to confirm the value.

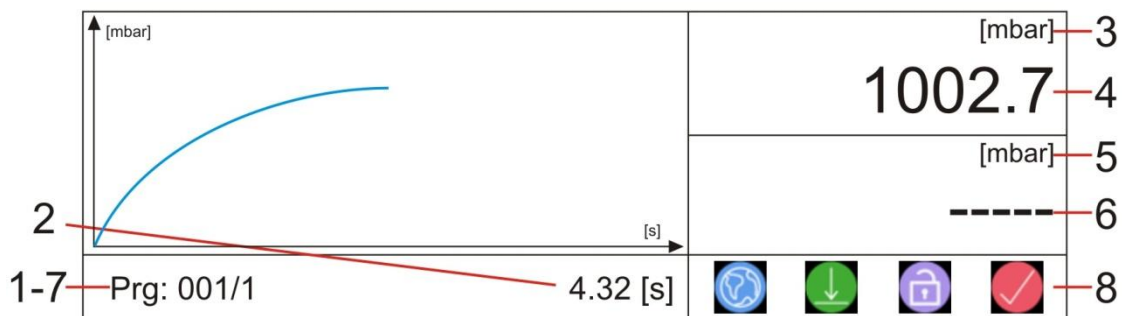
Instrument under test with graphic off:



In the field 1 it is indicated the current phase of the test (in English, whatever it is the language you have set): under it, you see on the left the field 7, that describes the current program, and on the right the time 2 to the end of the phase, with its relative unit. On the right you will find the indication of what the equipment is measuring: there is a high field 3-4 with its relative unit and numeric value, and a low field 5-6 with the same information (but referring to another measure) understood as pressure drop or difference between reference and test piece.

According to the phase where the equipment is (and according the calibration parameters and some test parameters), these fields display different measures: then we send you to the chapter Test Cycle. In the fields 4 and 6 a message “OVER” could appear: it identifies that the transducer relating to the indicated measure is full scale. In the field 6 dashes “-----” can appear in case that the test has not reached yet a phase in which you measure the relative dimension (for example in a filling phase of an absolute decay leak tester, Tx99x, do not measure the decay yet).

Instrument under test with graphic on:



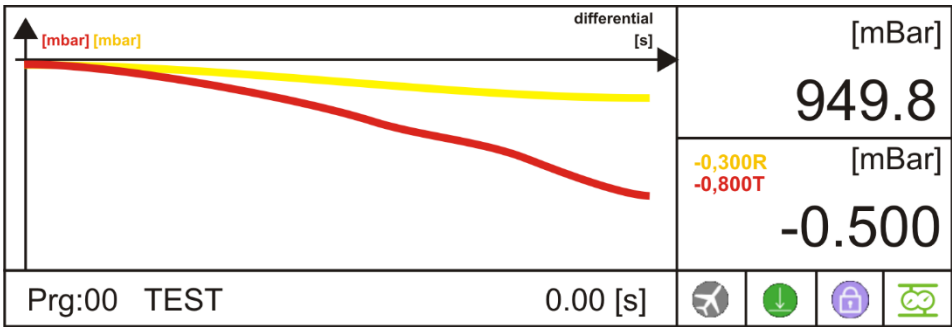
Compared to the case with graphic off, in the central-left field it is shown a graphic about the testing trend: the graphic and the measure/s that it shows are different according to the testing type that you are making and also to the phase in which the equipment is: therefore we send

you to the chapter Testing Cycle. The phase moves below beside the current program, while the remaining information are the same ones as in the case of graphic off.

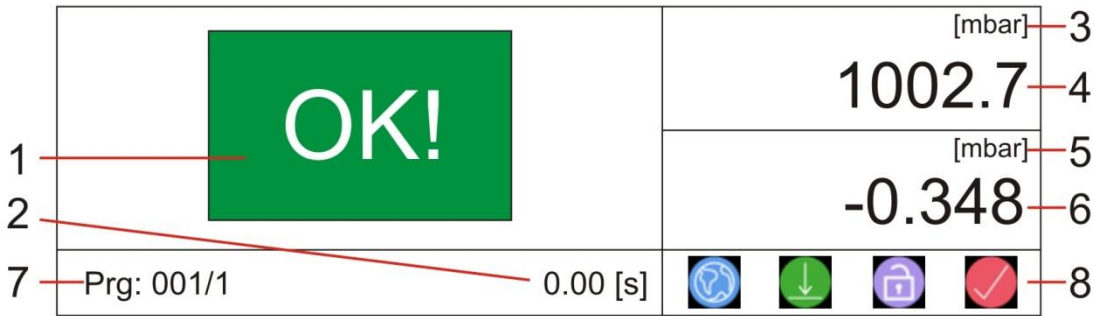
NB: in case of quick tests (very short total testing times and depending from the equipment model), the equipment disables automatically the graphic and some displays, in order to be very quick during the elaboration of the signals coming from the transducers.

In Differential and Dual absolute modes, the loss trend is represented simultaneously for both the reference part (yellow curve) and the test part (red curve), generally in scale with the loss value programmed for the test in progress.

The numerical loss values of the reference and test port part are also shown on the display, respecting the colors of the curves, as shown in the following screenshot:



Instrument with flashing result with graphic off:



The layout is the same as that one of the cases of the equipment under test with graphic off. The field 1 that was first used for the phase, now it visualizes the last phase made by the equipment, and alternated to it the result flashes (see below the possible results). Program number, field 7, is that one of just completed phase.

The field 2 visualizes the time remaining to the end of the phase, in the moment in which the test has been interrupt. On the right, in the fields 3-4-5-6 there are visualized the characteristic values of the test stored by the equipment (useful for reports) that depend on the phase in which the result has been generated (for more information see the chapter Test Cycle), and also by the equipment model (according to the model, they could be scrolling values if more of two ones).

In fields 4 and 6 it could appear at display, flashing with numeric values, some strings, that supply some more details about the kind of result:

- **"OVER"** (for the fields 4 and 6) to indicate that the test is finished because the full scale has been reached.
- **"P% Lo"** (for the field 4) to indicate that the test is finished because the pressure in the piece has decreased under the tolerance limit, or it has never reached the tolerance range (see the relative parameter of the chapter *Test Menu*)
- **"P% Hi"** (for the field 4) to indicate that the test is finished because the pressure in the piece has risen over the tolerance, or it has never reached the tolerance range (see the relative parameter of the chapter *Test Menu*).
- **"No Lev"** (for the field 4) to indicate that the test is finished because the pressure has never reached the set value (see the relative parameter of the chapter *Test Menu*).
- **P% LoR"**(for the field 4) to indicate that the test has ended because the pressure in the reference part has fallen below the tolerance, or has never reached the tolerance range (see the relevant parameters in the Test Menu chapter).
- **P% HiR"**(for the field 4) to indicate that the test has ended because the pressure in the reference part has risen beyond the tolerance, or has never reached the tolerance range (see the relevant parameters in the TestMenu chapter).

Possible Result Messages:

OK

NO OK

POS LEAK

KO REF

KO TOT

“OK”: good result

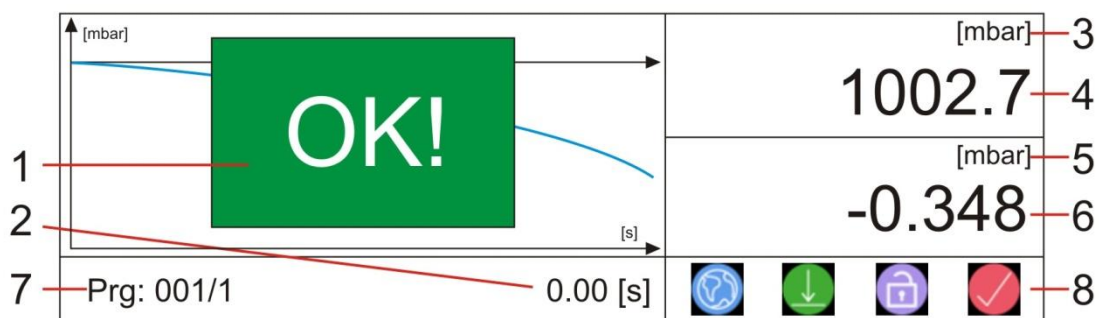
“NO OK”: waste result

“POS LEAK!”: positive decay

"KO REF": Reference side reject

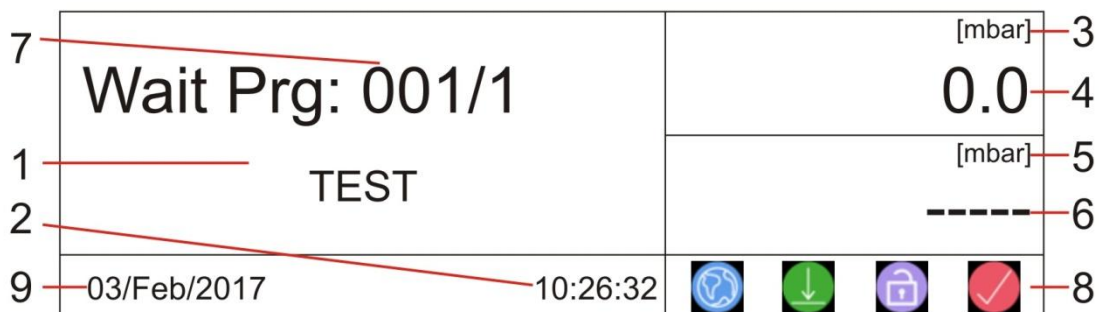
"KO TOT": Discard on both channels.

Instrument with flashing result with graphic on:



Compared to the case with graphic off, in the central-left field it is shown a graphic on the testing trend and alternating with it the result flashes. The last phase made by the equipment moves below beside the current program, while the remaining information are the same ones as in the case of graphic off.

Instrument in wait state:



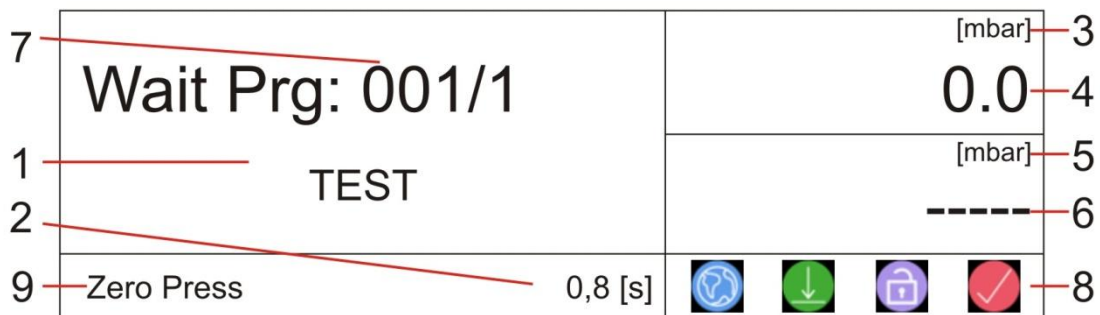
The equipment is waiting when it is not inside a menu, is not under test, does not display a flashing result.

There are different possible messages concerning the field 1:

- **"ERROR FLASH"**: potentially serious error, if this message appears you have to call right away the assistance
- **"ERROR EEPROM"**: warning, in case it happens, try to modify a test parameter and verify if the message continues to appear: if it does not appear anymore warning will be reset. In any case please contact the assistance.
- **"PR Full Scale!"**: the primary pressure transducer it is on the full scale: disconnect alimentation air pressure and possibly the piece, paying attention.
- **"VOUT Full Scale!"**: the secondary transducer it is on the full scale: disconnect alimentation air pressure and possibly the piece, paying attention.
- **"ERROR EVOUT"**: warning of power output: disconnect what it was connected to electric connector on the back panel (see the chapter Electric Components Back Panel), please contact the assistance.
- **"ERROR ELREG "**: warning of the electronic regulator: it is show when the electronic regulator doesn't succeed to set the wanted pressure, generally simply because the compressed supply air connected on the back panel is not sufficient or not present.
- **"WAIT Prg:"**: the equipment is ready for the start of the visualized program.

In any case below, in the fields 2 and 9, we find the date and the hour (format hh:mm:ss and dd/mm/yyyy where the month is in letters), while on the right, the fields 3 4 5 6, the same parameters that we find when a test is running.

Instrumentin autozero phase:

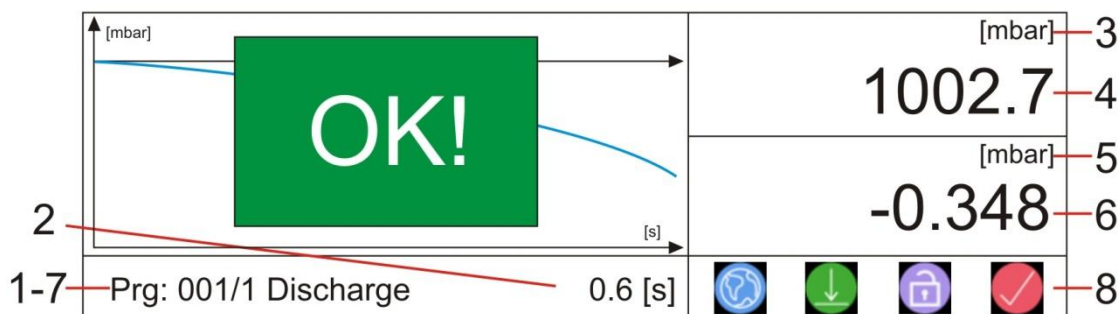


The visualization of autozero phase, field 9, current and of the relative time, field 2, is always on the line below on the left, both that it happens during waiting phase and with flashing result. The autozero has two possible phases:

- **"ZERO PRESS"**: it indicates that the pressure autozero is running
- **"ERR ZERO PRESS"**: it indicates that pressure autozero has not happened because an error has occurred,

The remaining fields, 3 4 5 and 6, are similar to the case of the equipment in waiting phase.

Instrument in discharge phase:

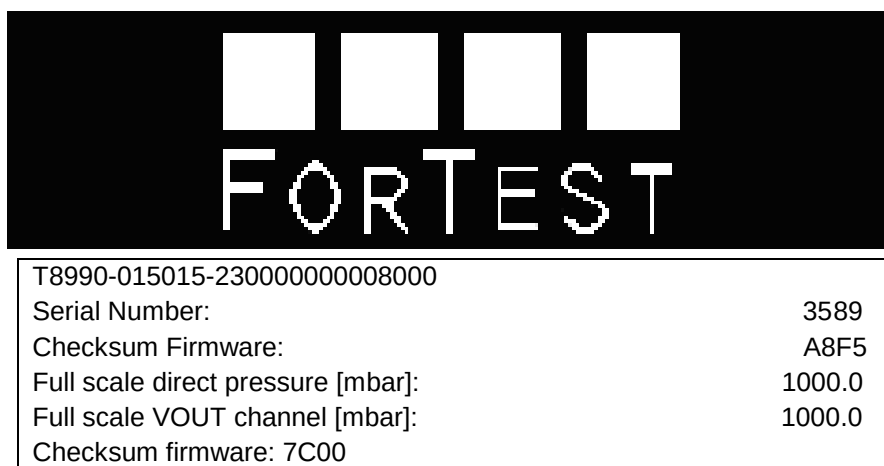


The display of the discharge phase, field 1, current and of the relative time, field 2, is always on the line below on the left, both that it happens in waiting phase and with flashing result. The discharge has only a possible phase:

- **“DISCHARGE”**: it indicates the discharge phase is running.

The remaining fields, 3 4 5 and 6, are similar to the equipment in waiting phase if the discharge follows an abort, otherwise they are similar to the case of the equipment with flashing result with graphic off.

Instrument at the start up:



At the start up the display shows for one second the company logo, and for the following two seconds some information similar to that ones that appears in the chapter Version Menu, that is the specific characteristics of the equipment. The information is (for a more detailed explanation we send you to the Version Menu chapter):

- Instrument code
- **“Serial Number:”**, serial number

- **“Checksum Firmware:”**, checksum of the firmware
- **“Full scale direct pressure:”**, direct pressure full scale and relative unit
- **“ Full scale VOUT channel:”**, channel VOUT full scale and relative unit.
- **“Checksum firmware:”**, display firmware checksums.

Test Cycle

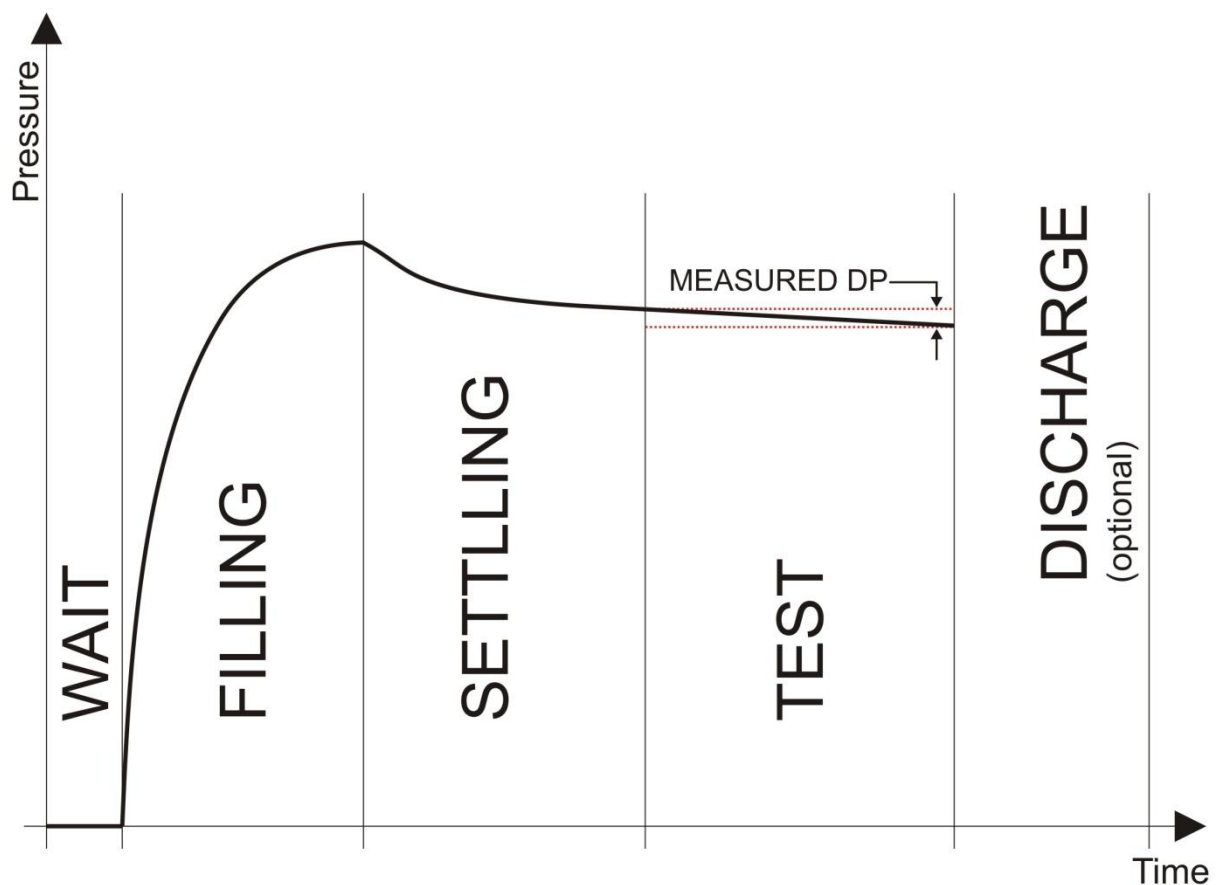
The equipment has like primary measure the direct pressure referred to the ambient one, and for second measure the pressure decay inside the piece under test.

Pressure measure will have a positive sign (sign not at display) in case of relative pressure higher than the ambient one, and a negative sign in the opposite case (depression).

A pressure decay corresponds to a pressure in the piece that decreases (in case of positive direct pressure, otherwise pressure decay will be in reality a pressure that will increase in case of test in depression), and for convention we display it with the sign “-“ if the kind of measure is in pressure, and with the sign “+” if the kind of measure is in pressure/time or volume/time (signs will be inverted in case of test in depression).

ABSOLUTE PRESSURE DECAY LEAK TEST CYCLE

A leak test is a kind of test in which you check if the piece is airtight, by putting inside this one a certain pressure and measuring in some way that after you fill it you have not too big pressure decay. Here below we show in a schematic way a standard leak test cycle:



Graphic shows the essential phases of leak test cycle, writing in the graphic the pressure inside the piece. Instead, below you will find the description of all the leak test phases made by the equipment, with the corresponding written message of each phase.

“Wait”/ “WAIT”: the equipment is waiting for start input.

“Filling”/ “FILLING”: This one corresponds to the filling phase, in which the inside pneumatic circuit work in order to allow the pressurization of the piece under test at the filling pressure you have set (see chapter *Test Menu*).

Filling phase can be made into two different ways, by time or by level: both modes are explained in chapter *Test Menu*, where relative test parameters can be set.

If the equipment in this phase does not succeed in filling in the right way according to parameters that have been set, then it will give waste as result. If the equipment does not succeed in filling piece under test (or if it fills it at a pressure that is higher than its limit), this could be caused to the piece itself (great leakage, wrong pneumatic connection, great elasticity of the piece), or simply caused by a low regulation of pressure regulator placed on frontal panel (see *Frontal Panel* chapter). In this last case, you have to set the pressure by the regulator, rotating the knob according to the directions marked on the same one in order to increase or decrease filling pressure (with electronic regulator optional, pressure will be set automatically by the equipment, then in this case setting has not to be made by the Customer).

During this phase (in reality already beginning from waiting phase) the equipment displays the pressure value inside the piece, this value is also coded on superior blue led bar in a proportional way regarding to equipment full scale (in case of equipment in depression, it will be proportional regarding to -1atm).

“Settling”/ “SETTLING”: This is the phase of adjustment, perhaps the most important one, that allows the stabilization of pressure and of temperature inside the piece under test. A too short time, in fact, will false the results of the entire test, causing for example false waste and false good result. In this phase the pneumatic circuit does not fill the piece anymore.

If during this phase pressure goes out of the tolerance range (whatever it is the filling type that has been set), then the equipment gives waste as result. In case that the settling tentative are activated (see the relative parameter in the chapter *Setup Menu*), the equipment, before of giving waste as result, will restart the phase filling for several times (always and only the filling, even if prefilling is activated) before of giving waste as result, a number of time according to the relative parameter.

At the end of this phase, direct pressure inside the piece visualization (both as numeric value on the display and as blue led number on the superior bar) will remain the same, because it identifies the real pressure in which test phase begins. In case of settling tentative, if the graphic is activated, it will reset the trace concerning the filling and the settling, restarting with a new graphic.

“Test” / ”TEST”: This is the phase in which the equipment makes the real test, that one that will recognize based on pressure leakage (case of test in pressure) inside the piece (measured during this phase) if it is hermetic enough or not, showing at the end of the test the result. The end of this phase is established by the relative setup parameter (**“IfBad:” / “At a bad outcome:”**), we send you, therefore, to the relative chapter *Setup Menu*. It will happen that pressure will increase instead of decreasing (case of test in pressure, the opposite one in depression): this could be caused by temperature phenomena or by wrong test parameters, and in this case the equipment will give a different result (**“OK”** or **“POS LEAK!”**) according to the entity of this increasing based on the parameter **“OppDecay:” / “Tolerance on the opposite decay:”**, see the chapter *Setup Menu*.

In this phase the equipment shows the pressure leakage inside the piece, as a numeric value in the appropriate field, and as numbers of led on (proportional according to higher leakage set in test menu, see the chapter *Test Menu*) in the bar below. If pressure leakage is not displayed as a pressure value but as something on a time, then you can see also red and green led on (at low light) to indicate a partial result.

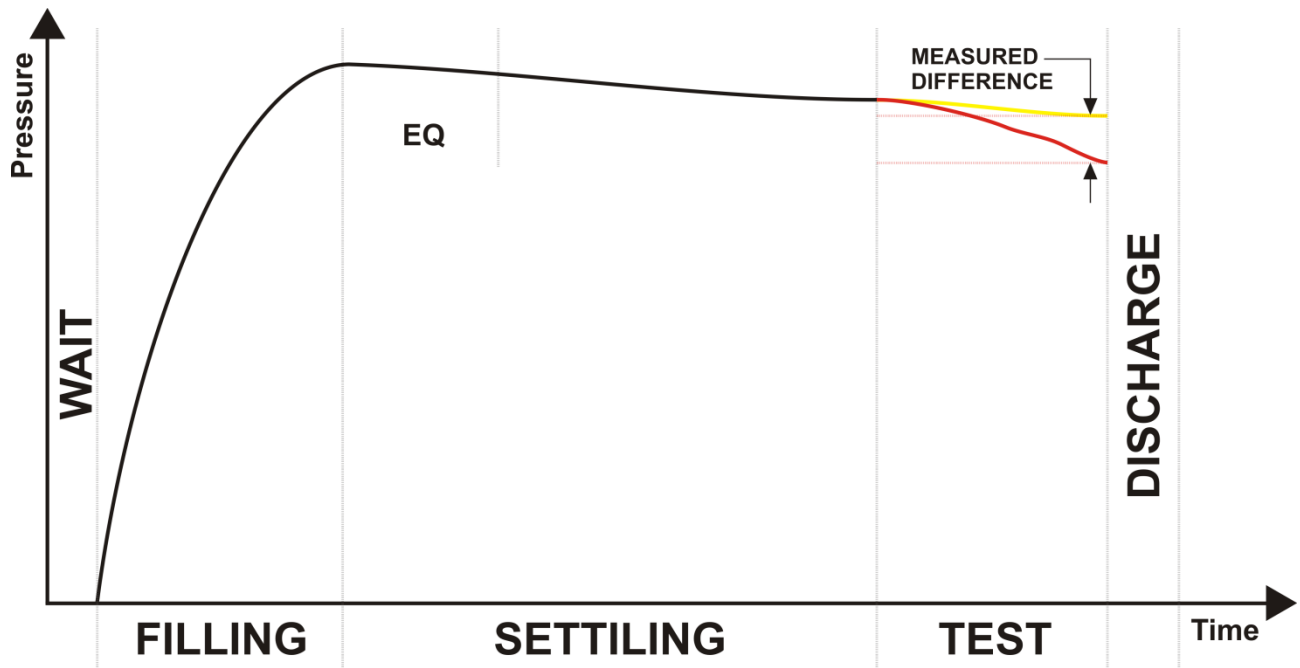
“Discharge”/ ”DISCHARGE”: in this phase the equipment empties the piece under test, obviously only if there is the discharge solenoid optional (internal or external). It is highly recommended for high test pressure, in order to avoid danger for the operator. Discharge phase, if activated, is made at the end of the test whatever it will be the way it ends: good result, waste result, any other result, abort.

The equipment in this case visualizes that discharge phase is arriving, but it continues to visualize both on the display and on the led the same information of the result phase; in case of abort it displays the same information as during wait phase.

DIFFERENTIAL DECAY LEAK TEST CYCLE

This kind of work allows to perform the leak test by detecting the pressure difference between a reference master piece and the piece to be tested.

The graph of the test cycle in differential mode is shown below:



“Wait” / “WAIT”: the instrument is waiting for a start input.

“Filling” / “FILLING”: This is the filling phase, in which the internal pneumatic circuit is activated to allow the pressurisation of the part under test and the reference part at the set filling pressure.

The filling of the parts can be done in two distinct ways, time or level: both modes are explained in the Test Menu chapter, where you can set the relative parameters.

If the instrument at this phase fails to fill the test piece or reference piece in accordance with the set parameters, then it will give a reject result. If the instrument fails to fill the piece under test or the reference piece (or fills them at a pressure higher than the threshold), this could be due either to the piece itself (large leak, incorrect pneumatic connection, strong elasticity of the piece) or simply to an insufficient adjustment of the pressure regulator located on the front panel (see Front Panel chapter). In the latter case, adjust the pressure through it, turning the knob according to the verses marked on it to increase or decrease the filling pressure (with the optional electronic regulator the pressure is automatically adjusted by the instrument, so this adjustment has not to be done by the customer).

During this phase the instrument shows on the display the pressure value inside the piece, a value coded also on the upper blue led bar in a proportional way with respect to the instrument bottom scale (in case of vacuum instrument, it will be proportional with respect to -1atm).

“Settling” / “SETTLING”: This is the settling phase, perhaps the most important, which allows the stabilization of pressure and temperature within the part under test. A too short time in fact would distort the results of the entire test, causing for example false rejects or false good. For a programmable time in the parameter "Equalisation in settling" a percentage is set on this phase (initial part of the settling phase) in which the Reference part and the test part are in communication, in order to perform a better balancing of the pressures in the two parts before entering the test phase, where the pressure difference between the two parts is detected.

At the end of this part of the settling phase the pressures inside the two pieces are isolated from the pneumatic circuit and the pressures are stabilized before reaching the leakage measurement phase.

"Test" / "TEST": this is the phase in which the instrument performs the test, the one that will determine the result of the test on the basis of the pressure difference between the test part and the reference part.

The end of the phase is determined by the relevant setup parameter ("**IfRej:**" / "**At a bad outcome:**"), please refer to the Setup Menu chapter. It may happen that the pressure inside the piece, instead of decreasing, increases compared to the reference piece (in the case of a test under pressure, in depression vice versa): this may be due to thermal phenomena or incorrect test parameters or a leak in the reference piece. In this case the instrument will give as result "NO OK REF" to indicate an anomaly compared to the reference side.

"Discharge" / "DISCHARGE": In this phase the instrument empties both the part under test and the reference part, obviously only if the optional exhaust valve (internal or external) is present. It is strongly recommended for high test pressures, in order to avoid dangers for the operator. The discharge phase, if enabled, is performed at the end of the test whatever the way it ends: good result, reject result, any other result, abort. In this case the instrument shows that the discharge phase is taking place, but it continues to show both on the display and on the leds the same information of the test phase; in case of abort it shows the same information as in the waiting phase.

DUAL ABSOLUTE LEAK TEST

In the same way as the differential test cycle, this mode of operation allows to detect a leak as the difference between a reference piece and a test piece.

Unlike the pure differential mode, however, the dual absolute mode allows the reference part to be filled at programmable intervals and at the same time to acquire the pressure trend during the test phase..

In this way it is possible to obtain a pressure difference no longer directly between reference piece and test piece but between a pressure trend stored inside the instrument itself and the test piece. At regular intervals, it is performed a sampling of the trend in the reference piece, in order to update the values.

This allows a series of advantages over the classic differential system illustrated above such as the tracking of the variation of the ambient temperature over time, less or almost no mechanical stress on the reference part, less or almost no thermal stress on the reference part. All this leads to having differential tests always at the ideal reference conditions with respect to the part under test.

For the test cycle, please refer to the differential case. The only difference is the one indicated above and that is that, through the parameter (found in the test menu): "Freq. acquire reference", it is possible to define the sampling time of the reference.

N.B. In dual absolute mode, settling equalization is not active.

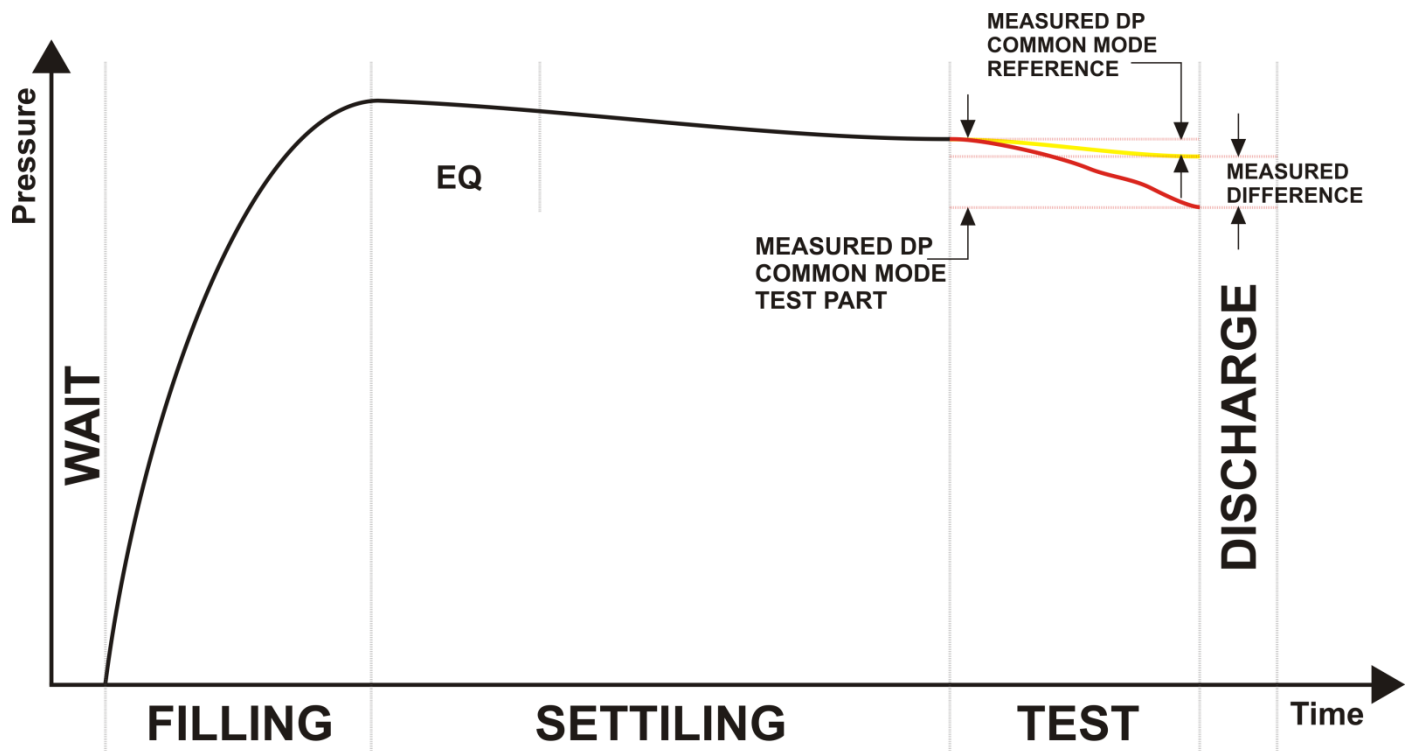
ZERO CENTER LEAK TEST

This mode of operation allows the leak test to be performed on two pieces at the same time.

Two pieces are then connected to the instrument at the same time, one on the test side and one on the reference side. On the reference side it is no longer mandatory to connect a real reference piece because the instrument will consider it as a second piece to be tested.

Please refer below where we talk about the test phase, how it is decided if a part is good or waste.

The test cycle is represented by the following graph:



“Wait” / “WAIT”: the instrument is waiting for a start input.

“Filling” / “FILLING”: This is the filling phase, in which the internal pneumatic circuit is activated to allow the pressurisation of the part under test and the reference part at the set filling pressure.

The filling of the parts can be done in two distinct ways, time or level: both modes are explained in the Test Menu chapter, where you can set the relative parameters.

If the instrument at this phase fails to fill the parts in accordance with the set parameters, then it will give a reject result. If the instrument fails to fill the piece under test or the reference piece (or fills them at a pressure higher than the threshold), this could be due either to the piece itself (large leak, incorrect pneumatic connection, strong elasticity of the piece) or simply to an insufficient adjustment of the pressure regulator located on the front panel (see Front Panel chapter). In the latter case, adjust the pressure through it, turning the knob according to the verses marked on it to increase or

decrease the filling pressure (with the optional electronic regulator the pressure is automatically adjusted by the instrument, so this adjustment has not to be done by the customer).

During this phase the instrument shows on the display the pressure value inside the piece, a value coded also on the upper blue led bar in a proportional way with respect to the instrument bottom scale (in case of vacuum instrument, it will be proportional with respect to -1atm).

"Settling" / "SETTLING": This is the settling phase, perhaps the most important, which allows the stabilization of pressure and temperature within the part under test. A too short time in fact would distort the results of the entire test, causing for example false rejects or false good. For a programmable time in the parameter **"Equalisation in settling"** a percentage is set on this phase (initial part of the settling phase) in which both parts are in communication, in order to perform a better balancing of the pressures in the two parts before entering the test phase, where a leakage will be detected.

At the end of this part of the settling phase the pressures inside the two pieces are isolated from the pneumatic circuit and the pressures are stabilized before reaching the leakage measurement phase.

"Test" / "TEST": this is the phase in which the instrument performs the test, the one that will determine the result of the test on the basis of the loss of pressure inside the part, showing the result at the end.

The end of the phase is determined by the relevant setup parameter (**"IfRej:" / "At a bad outcome:"**), please refer to the Setup Menu chapter. It may happen that the pressure inside the piece, instead of decreasing, increases compared to the reference piece (in the case of a test under pressure, in depression vice versa): this may be due to thermal phenomena or incorrect test parameters or a leak in the reference piece. In this case the instrument will give as result "NO OK REF" to indicate an anomaly compared to the reference side.

During the test phase, the instrument will determine whether the parts are good or defective based on the differential loss detected (loss intended as the difference between the part on the reference side and the part on the test side). The acceptable differential loss threshold is defined by the "Maximum loss" parameter in the test menu.

Using the particular dual absolute hardware, however, it is also possible to define a further common absolute pressure drop of the 2 channels; the parameter (present in the test menu) is: "Trigger common mode". This allows you to intercept any anomalies in the test such as for example 2 pieces with similar leakages.

In this case, the instrument performs a re-test of the two parts (it re-tests only the test phase) for a time set by the parameter: **"Zero-center re-test duration"**. During this time, the instrument will no longer perform a differential test but will only show the absolute drop of the two channels, taking as maximum leakage the one set by the parameter **"Maximum decay"**..

You can explain how this mode works with this example.

Suppose we want to discard parts under test with a sensitivity of 0.5 cc/min (as differential) but taking as reference a good part in absolute mode a maximum loss of 1cc/min.

In this case the parameters have to be set as follows:

Maximum decay: 0.5cc/min

Trigger common mode: 1cc/min

In this case the following scenarios may occur:

Both pieces have a lower loss than the common mode loss.

Example:

Test: 0.6 cc/min

Reference: 0.7 cc/min

The result will give a differential of 0.1cc/min (i.e. lower than the maximum set threshold) and each drop is less than the common mode loss. The instrument will give as result: good on both channels.

Example of loss:

Test: 0.8 cc/min

Reference: 0.2 cc/min

The differential will indicate a loss of 0.6 cc/min on the test side (but in absolute decay, less than the trigger common mode) and then give the result: Reject on the test side but good the test side..

There may be a case where a channel has a higher loss in a common way.

Example:

Test: 0.8 cc/min

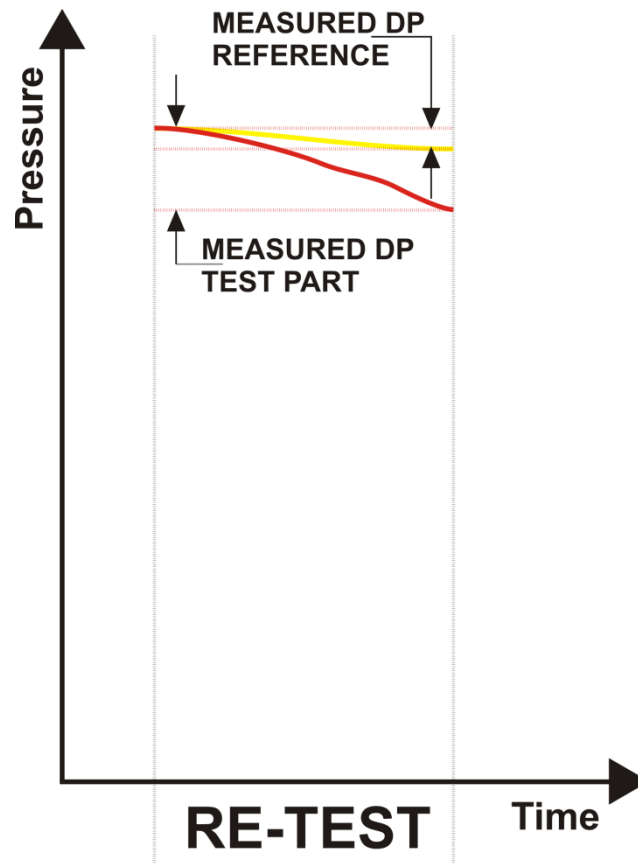
Reference: 1.5 cc/min

In this case the instrument performs a re-test in order to check the correct tightness of the two pieces and then shows the result.

You can see a graphical representation of normal and re-test cases in the graphs above and below. In the test case without re-test the "Trigger common mode" thresholds are indicated as "Measured dp common mode" and "Maximum drop" is indicated as "measured difference". In case of re-test the two thresholds are indicated as "measured dp difference".

"Discharge" / "DISCHARGE": In this phase the instrument empties both the part under test and the reference part, obviously only if the optional exhaust valve (internal or external) is present. It is strongly recommended for high test pressures, in order to avoid dangers for the operator. The

discharge phase, if enabled, is performed at the end of the test whatever the way it ends: good result, reject result, any other result, abort. In this case the instrument shows that the discharge phase is taking place, but it continues to show both on the display and on the leds the same information of the test phase; in case of abort it shows the same information as in the waiting phase.



“Re-Test” / “RE-TEST”: this phase is performed only if the pressure drop detected in one of the two pieces (or both) is higher than the Trigger common mode parameter (see chapter Test menu). The duration of this phase is defined by the time set in the zero-center re-test duration parameter (see Test menu chapter).

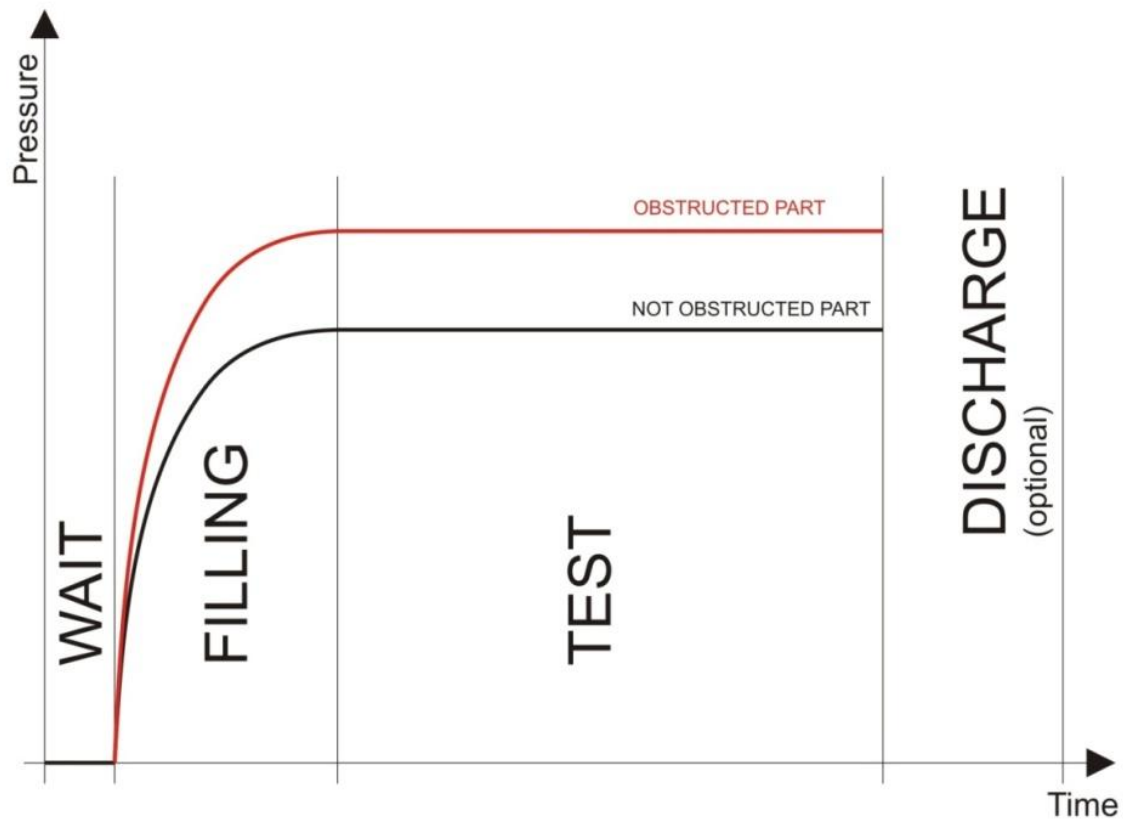
During this phase the instrument checks that both pieces have a pressure drop lower than the maximum set drop. If a greater pressure drop is detected on one or both of the two pieces under test, the instrument will provide a rejection result.

OBSTRUCTION TEST CYCLE

Obstruction test allows to verify in an approximate way if a piece is obstructed or to verify its flow capacity. What you have to make it is to open the air to fill the piece, and verify in testing phase that

the pressure you have measured remains inside the range fixed by the test parameter (see the chapter *Test Menu*).

Phases that compose obstruction test made by the equipment are almost the same as the ones of leak test.



Differences are:

- There is not any settling phase
- Transaction from filling phase to test phase is underlined by a beep
- On the graphic there is always and only direct pressure.
- During the testing phase you continue to put air into the piece.
- As this kind of pressure measures only direct pressure, during the testing phase you continue to visualize both on the display and on the led this measure, leaving the field concerning the leakage and the inferior bar of led empty

Test Menu

Test menu is the menu that contains, indexed at the program number, the test parameters of each single program: these are the parameters that characterize the test that the equipment makes (for more details, see the chapter *Test Cycle*). In all the screens it is visualized the program number actually under modification, on the top to the right the arrows navigation: only in the first screen it is possible to select and to modify the program number (as any other normal parameter). If you modify the program number, the equipment uploads and visualizes the stored parameters for that program. When you exit from the menu, the last selected program, will be then the activated one.

To enter in this menu is enough press **[TST]** button on the frontal panel of equipment. To exit of the menu, press again **[TST]** button to come back in waiting status. Press **[SET]** button to exit from test menu and directly accede to Setup menu.

Here below the menu screens are visualized in order of appearance into the display (from the first one to the last one) and its parameters are described: the chapter refers to the standard menu, that is without any optional activated. Some optional activate other sub menu, activate other parameters, modify the meaning of some standard parameters.

All these modifications will be discussed in the chapter concerning the optional itself. Moreover, the following screens and parameters refer to all T8000 instruments: according to the single model, you can have some modifications, that in the case are always indicated. In case that a parameter cannot be modified or selected, at the place of its value it will appear dashes "----".

First screen of TEST menu:

TEST MENU: FILLING, PRG:	001/1	▲	▼
Kind of test:	Decay	[----]	
Filling pressure:	1000.0	[mbar]	
Pressure tolerance:	20	[%]	
Kind of filling:	Time	[----]	
Program name: TEST			 

In this screen it is possible to modify the program number, that will appear on the first line besides the arrows.

The number after the "/" indicates the program group in which they are linked selected programs. In this way, you can also call up groups of programs directly from the external signal booster programs. For example, if you chain 5 programs which will form the group "3", simply clicking through AUX1 signals the START 3 INPUT, the instrument will start the test from the first program of the group "3", and NOT the program 3.

In addition, one of the possible operating modes for the DUAL ABSOLUTE T8090 can be set via the "**Kind of test**" parameter.

"Kind of test:" this parameter allows to select the test type to be made in this program. It can have the value:

- **"Zero-Center"**: it identifies that this program is related to a test where two parts (which must be identical) are tested at a time, taking advantage of the difference in the pressure measurement during the test phase.
- **"Decay"**: it identifies that this program is concerning to an absolute pressure decay leak test using the TEST SIDE only connected to a piece under test.
- **"Differential"**: it identifies that this program is related to a standard differential leak test, detecting the difference between the test piece and the reference master piece.
- **"Obstruction"**: it identifies that this program is concerning to an obstruction test.
- **"Dual Absolute"**: it identifies that this program is related to a differential leak test by detecting the difference between the test piece and the reference master piece, sampling the Reference values periodically in order to obtain a pressure behaviour similar to the test piece optimizing cycle times and test performance.
- In case of firmware customize, the parameter allows to select different test cycles, not put in standard versions.

According to the type of selected test, the screen concerning the test phase change (see here below).

"Filling pressure:" this value expresses the pressure with which the piece under test will be filled. If the equipment has a negative pressure full scale, or it is equipped for positive and negative measure, it will be possible to set up the negative filling pressure keeping pressing the "0/-" button on the graphic keyboard that will appear when this parameter has been selected.

In case of level filling (**"Kind of filling" = "Level"**), if you do not reach the limit set during the time **"Filling phase duration:"**, the result will be waste. In case of filling time (**"Kind of filling" = "Time"**), the equipment controls that the pressure at the end of this time is included in the range **"Filling pressure" +/- "Pressure tolerance:"** (filling pressure +/- filling tolerance); if the measured pressure is over the set range, the result will be waste. Whatever it will be the filling time you have set, if during the adjustment phase the pressure goes over certain limit (see *Setup Menu* chapter), then the equipment will give a waste result. In case you are making an obstruction test, this parameter will lose the meaning, because the filling is set by minimum and maximum test pressures (see after).

“Pressure tolerance:” percentage of the pressure value at which the test is allowed: if, at the end of any phase of the test, it is detected a pressure value outside this tolerance, then the instrument will give a reject result.

“Kind of filling:” filling type: it indicates the modality to fill the piece under test; if it is set at the level (**“Level”**), the equipment, to pass at the settling phase should reach the set pressure in the field **“Filling pressure”** within the time **“Filling phase duration:”**, otherwise it will stop the test giving waste result.

In "Dual Absolute", "Differential" and "Zero Center" modes, the level is reached when both channels detect a pressure equal to or greater than the set filling pressure.

This mode is not available if the instrument is equipped with the optional electronic regulator.

If it is set in time (**“Time”**), the equipment fills during all the time **“Filling phase duration:”** and at the end of this phase it checks that the pressure is in the range **“Filling pressure”** +/- **“Pressure tolerance”**. in negative case, the equipment stops the test living waste result.

Also in this case in the "Dual Absolute", "Differential" and "Zero Center" modes it is defined as correct filling time when both channels detect a pressure within the pressure set +/- the value set in the parameter "Pressure tolerance".

In case you're making an obstruction test, filling is always managed and locked by time.

NB: this parameter becomes **“Filling attempts”** if the electronic regulator is set and activated at software as an optional.

Second screen of TEST menu:

TEST MENU: PHASES TIME, PRG:	001/1	▲	▼
Filling phase duration:	10.0	[s]	
Settling phase duration:	20.0	[s]	
Test phase duration:	5.0	[s]	
Discharge phase duration:	1.0	[s]	
Program name: TEST			 

“Filling phase duration:” this parameter represents the total or maximum duration of the filling phase, according to the setting of the filling type.

“Settling phase duration:” it indicates the adjustment time (stabilization) of the piece under test and of the piece connected to the REFERENCE branch, in case the test type is set as "Dual Absolute", "Differential" and "Zero Center".

NB: in case of an obstruction test, this parameter is blocked to zero, because this phase is not present in the cycle test.

“**Test phase duration:**” it indicates the test time of the piece under test.

“**Discharge phase duration:**” it indicates the air discharge phase at the end of the test. Obviously, the discharge will work only if the equipment is supplied inside or outside of a discharge solenoid (that can be an optional or not according to the type of equipment).

The discharge acts on both test channels in all the kind of tests in which it can be used.

Third screen of TEST menu

TEST MENU: DECAY, PRG:	001/1	▲	▼
Kind of measurement:	Pressure	[----]	
Maximum decay:	-0.500	[mbar]	
Piece's volume:	0.0	[cc]	
Offset on the decay:	0.0	[mbar]	
Program name: TEST			 

“**Kind of measurement:**” measure type: it indicates how you want to visualize the measure of the pressure decay inside the piece under test. Modes are the following ones:

- “**Pressure**”: the equipment visualizes the leak like a pressure, in the equipment unit measure (set in the Calibration Menu, parameter “**Unit of measure VOUT**”). In this mode the equipment visualizes pressure decay with a number of decimal points established by the parameter “**Decimal points VOUT channel:**” placed in the Calibration Menu.
- “**Pressure/time**” pressure/time: the equipment visualizes the leak like a pressure according to a time, where the pressure is in the equipment unit measure (set in the Calibration Menu, parameter “**Unit of measure VOUT:**”), and the time is in seconds. In this mode the equipment visualizes the pressure/time with a number of decimal points established by the parameter “**Decimal points VOUT channel:**” placed in the Calibration Menu, plus 1.
- “**cc/h**” cc/h: the equipment visualizes the leak in this unit measure. In this mode the equipment visualizes with one decimal point.
- “**cc/min**” cc/min: the equipment visualizes the leak in this unit measure. In this mode the equipment visualizes with two decimal points.
- “**cc/s**” cc/s: the equipment visualizes the leak in this unit measure. In this mode the equipment visualizes with two decimal points.

- **“mm³/s”** mm³/s: the equipment visualizes the leak in this unit measure. In this mode the equipment visualizes with two decimal points.

NB: the two measures volume/time needs to set a testing volume different from 0. In case that the parameter of the volume has been set at 0, the equipment will force the kind of measure in the mode **“Pressure”**. In case the equipment enables the calculation of the volume measure (volumetric optional activated and also the volumetric test activated) and always with a volume set as null, then during the test the equipment will use a volume saved before, at the end of the leak test it will concatenate the volumetric test for the volume calculation (of the piece or of the cavity of the bell if it is a Mx98x equipment), based on this new volume it will calculate again the leak (and it will update the saved volume).

“Maximum leak:” maximum leak: this is the maximum value of pressure leak allowed during the leak test phase: it will be a pressure decay inside the piece.

“Piece's volume:” volume under test: this parameter allows the conversion of pressure decay in a leak of volume/time type, according to the parameter set **“Kind of measurement”**. It must be set equal to the volume of the testing piece, it is necessary to count pipe fitting volume from the equipment to the volume under test, otherwise the conversion could take to wrong leak values.

“Offset on the decay:” offset on the decay: it allows to subtract or to add a value (an offset) to the measured leak. This parameter is useful to eliminate any adjustment queues that repeat always and all the same for the piece under test. Attention: a wrong use of this parameter can bring to give wrong results of measure because, if it is set at a too high value, it could mask a real leak. It has the same unit and the decimal points of the parameter **“Maximum leak:”**.

Third screen of TEST menu, only in case that “type of test” parameter is set as “Obstruction”

MENU TEST: OBSTRUCTION, PRG:	001/1	▲	▼
Min pres obstruction test:	800.0	[mbar]	
Max pres obstruction test:	1000.0	[mbar]	
Program name: TEST			 

“Min pres obstruction test:” minimal pressure of obstruction test: minimal pressure to which the valve (or the piece) submitted to an obstruction test must open. If the equipment, at the end of the testing phase measures a pressure lower than that one that has been set, it will give a result of waste.

“Max pres obstruction test:” maximal pressure obstruction test: maximal pressure at which the valve (or the piece) submitted to the obstruction test has to be opened. If the equipment, at the end of testing phase measures a pressure higher than the set one, it will give a waste result.

Fourth screen of TEST menu:

MENU TEST: OTHERS, PRG:	001/1	▲	▼
Prg following the running one:	No	[prg]	
Initial delay at start:	0.0	[s]	
Output coupled with the program:	----	[out]	
Input coupled with the program:	----	[in]	
Program name: TEST		   	

“Prg following the running one:” program following the current one: in case of good result, a start to the program set in this parameter has been given automatically. It is particularly easy to make complex cycles with more testing programs or little automations. We have to notice that at the end of the sequence of linked programs, a following generic start command will start again the first of the programs of the sequence.

Zero program cannot be linked, because if this parameter is at zero, this means that there are not any programs to link to it. A program can't link itself (that means that if this parameter is equal to the program that it contains it, the linked sequence will not be done).

In case of program linked, the AUX1 connector signals behaviour change. If the actual program ends with a waste (or even with an abort), and it there is a program linked, this last one does not run, and the entire test stops it: in this case, the signal behaviour is the same. If the actual program ends with a good piece, and there is a program linked, this last one will start in the first useful instant for a new start (then at the end both of the **“Label signal duration:”** time and the discharge time **“Discharge phase duration:”**). The AUX1 connector good signal remains activated for all the **“Label signal duration:”** time if it is different from zero, otherwise it does not activate at all. The **“End of cycle signal:”** signal remains activated during the linked sequence. All this described is valid for the parameter combination **“Bad management:”** = **“Standard”**, **“Abort management:”** = **“Standard”**, **“End of cycle signal:”** = **“Standard”**.

“Initial delay at start:” initial delay at the start: after having given the start, this timer runs, that will delay the departure of the filling phase (or prefilling), of an effective time that depends also on the parameter setting **“Input coupled with the program:”**.

“Output coupled with the program:” this parameter is activated only if the equipment has activated the expansion optional, and it assemble the same one. Briefly we can say that this function allows the activation of some outputs during the test, but we send you to the chapter of the Expansion I/O for more details.

“Input coupled with the program:” this parameter is active only if the equipment has activated the expansion optional and it assembles the same one. Briefly we can say that this function manages the testing start in the following way: if within the time **“Initial delay at start:”** the equipment does not see the selected inputs as activated, then give the abort to the test. We send to the chapter of the Expansion I/O optional for more details.

Fifth screen of TEST menu (Zero-Center mode only):

MENU TEST: DUAL PR:	001/1	▲	▼
Multip. Test time:	x 1.0	[----]	
Multip. Trigger Zero-Center:	x 2.0	[----]	
Zero-Center re-test duration:	10,0	[s]	
Triger common-mode:	-2.000	[mbar]	
Program name: TEST			 

“Multip. Test time:” This parameter sets the re-test duration as a multiplication of the test time. The final test time must guarantee, in absolute drop mode, the correct detection of the loss rate that would normally be detected in differential mode.

“Multip. Trigger Zero-Center:” This parameter sets the maximum acceptability threshold in absolute pressure drop mode on both channels. It must be set to ensure the correct detection of the leakage rate that would occur in case of absolute pressure drop measurement.

“Zero-Center re-test duration:” parameter not editable, but only for visualization. It represents the re-test time in case the loss detected on at least one of the two channels is greater than the "Trigger common mode".

“Trigger common-mode:” parameter not editable, but only for visualization. Represents the maximum permissible loss of a single channel. If the instrument detects a loss (in a single channel or in both) greater than this parameter it will repeat the test phase for the duration time Re-Test Zero-Center.

During the re-test the maximum value of loss allowed on both channels will be "maximum drop"; therefore the instrument will not perform tests in differential mode on the two pieces but only in absolute drop.

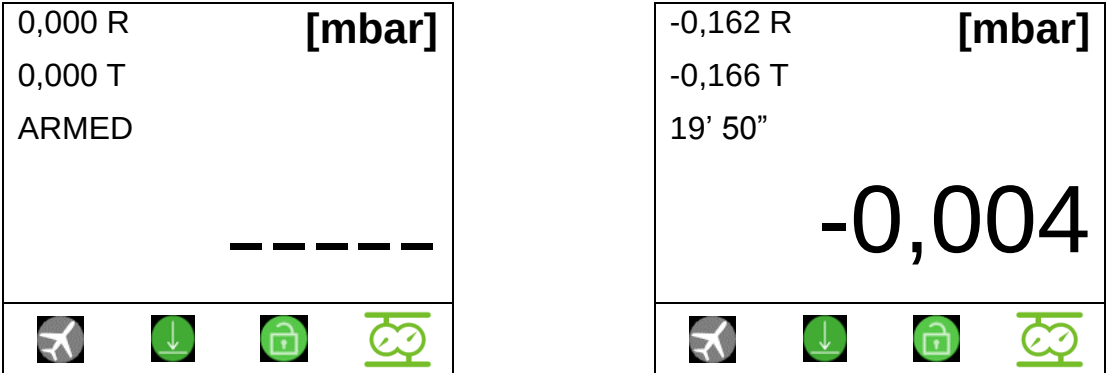
Fifth screen of TEST menu (Dual Absolute mode only):

MENU TEST: DUAL PR:	001/1	▲	▼
Freq. Campiona reference 20 [min]			
Program name: TEST			

“**Freq. acquire reference:**” this parameter indicates the time interval with which the pressure trend in the reference piece is sampled. This value will be saved and used as a basis to calculate the pressure difference with the piece under test and thus discriminate good pieces from bad pieces.

Until the next sampling, the Reference piece is left at ambient pressure to allow it to return to conditions similar to those on the pieces under test.

The display always shows when the reference sampling will be performed (ARMED flashing) and the missing time to the next acquisition of the value, as shown in the figure:



In the left screen the instrument is ready to acquire the reference piece, while in the right screen the instrument has acquired the reference piece (value in yellow, with letter R) and the test piece (value in red, with letter T) indicating as the final value of the test -0.004 or the difference between test port and reference piece.

Fifth screen of TEST menu:

TEST MENU: MEASURE, PRG:	001/1	▲	▼
Measure unit of pressure:	mbar	[----]	
Decay unit measure:	mbar	[----]	
Program name: TEST			  

“Measure unit of pressure:”: this parameter needs to change measure unit of filling pressure.

“Decay unit measure:”: this parameter needs to change measure unit of decay pressure. When you configure the equipment (through “type of measure” parameter) in “pressure” or “pressure/time”, through this parameter will be possible to configure measure unit.

Sixth screen of TEST menu:

TEST MENU: COPY, PRG:	001/1	▲	▼
Program to be overwritten:	----	[----]	
Confirm overwriting?	No	[----]	
Program name: TEST			  

“Program to be overwritten:” This parameter indicates to which program you want to copy the settings of the current program.

“Confirm overwriting?” This parameter is used to confirm the copy of the program. If it is set to “Yes”, the current program will be copied to the program set in the “**Program to be overwritten:**” parameter.

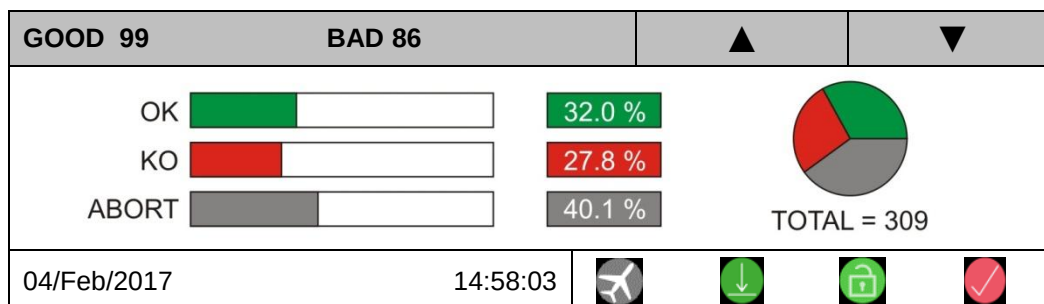
Count Pieces Menu

Count pieces Menu is the menu that contains the amount of the tests made by the equipment, of the pieces decreed as good ones and of the waste ones (these three counting can be put to zero.)

To enter in Count Pieces Menu is enough to press **[CTP]** button present on frontal panel of equipment. To exit from this menu and return to waiting status press again **[CTP]** button. To accede directly to Setup menu or Test menu press **[SET]** or **[TST]** button.

Here below the menu screens are visualized in order of appearance into the display (from the first one to the last one) and its parameters are described: the chapter refers to the standard menu, that is without any optional activated. Some optional activate other sub menu, activate other parameters, modify the meaning of some standard parameters. All these modifications will be discussed in the chapter concerning the optional itself. Moreover, the following screens and parameters refer to all the T Instruments: according to the single model, you can have some (few) modifications, that in the case are always indicated. In case that a parameter cannot be modified or selected, at the place of its value it will appear dashes “----”.

First Screen of Count Pieces Menu



This screen does not contain any parameters but it is only a visualization one.

“Good=”: number of pieces tested by the equipment that gives a good result from the last zero made. The meter counts all the good ones, then in case of the third result option activated and in case it is a leak tester Tx9xx (but exclude the differential leak tester Tx96x) or micro flow tester Tx71x and Tx72x, this meter is a sum of the good ones and of the good ones with reservation.

“Bad=”: number of pieces tested by the equipment that gives a result of waste from the last zero made. The meter counts all the waste pieces, then also the waste of the piece of reference, in case of differential leak testers Tx96x, the waste due to the bell test for compliance leak testers Tx98x, the waste due for flow under or up the limit for tester Tx73x and Tx74x (with the option of the third result activated)

Moreover, we supply a graphic-statistic visualization that includes also terminated tests through ABORT.

Second screen of Count Pieces Menu

BUONI 99		SCARTI 86		▲	▼
253	Abort	001	0.0 [mbar]	0.0 [mbar]	
252	Good	001	1002.3 [mbar]	-0.020 [mbar]	
251	Good	001	1001.7 [mbar]	-0.024 [mbar]	
250	Bad	001	995.4 [mbar]	-3.189 [mbar]	
04/Feb/2017		14:58:03			 

Here below on every one of the four lines there are at display the characteristic data of the results that the equipment has stored: here below there is a description of every field that composes each line, from the left to the right. At the end of each test (no matter how it ends) the equipment will save some characteristic data of the test itself, that are generally used for its storage: this memory is not lost when the equipment is off (as it happens for the goods and wastes counts).

Number result: the number of the result associated to the stored test on this line. The number of result is always the highest on the top, then it will decrease toward the downside: at a lower number, it corresponds an older result (that is the result “0” is the oldest one), that is the stored results are in temporal order from the newer (upside with a higher associated number) from the older one (below with a lower associated number). The equipment can store till 4000 results, that will be numbered from “0” to “3999”: in case the maximum limit of stored results will be exceeded, it is cancelled always the older one that is the “0”.

Result: the result associated to the stored test on this line. This one can be:

- **“Good”** in case of good: we mean all the good ones, than in case that the option of the third result is activated and it is a leak test equipment Tx9xx (excluded the differential Tx96x) or micro flow test Tx71x e Tx72x, this result is associated also to the good ones with reserve.
- **“Bad”** in case of waste: we mean all the waste ones, then also the waste of the reference piece in case of differential leak test instruments Tx96x, the waste due to the test under bell for compliance leak test instruments Tx98x, the waste limit for floe test instruments Tx73x and Tx74x (with the option third result activated)
- **“Abort”** in case of abort
- **“OppDec”** in case of inverted decay (see the parameter of Setup Menu **“Tolerance of inverse decay:”**)

Number of program: number of program associated to the stored test on this line.

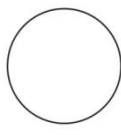
Stored pressure: represent the test pressure stored with the pieces has really filled.

Stored VOUT: represent the test VOUT stored.

When we are on the navigation arrows upside on the right, with the button **[E]** as in any other menu it is possible to enter inside the same one. Here there are not parameters that can be modified, but

when you are on a line, the number of the corresponding result will flash: pressing **[E]** it is possible to enter in a new menu that shows in a more detailed way the stored result (see here below). With the arrow buttons **[▲]** and **[▼]** it is possible to scroll the results, from the higher number till **"0"**.

In case there are not any stored results it will appear a screen of this kind:

GOOD 0		BAD 0		▲		▼	
OK		0.0 %		 TOTAL = 0			
KO		0.0 %					
ABORT		0.0 %					
04/Feb/2017		14:58:03					

If there are less than four stored results, there will be obviously some empty lines (that will be always in the lower part of the display).

In case of malfunction, or more easily if the count pieces have never been put at zero, it will be visualized instead of the previous screen the following one:

GOOD 0		BAD 0		▲		▼	
ATTENTION: Count pieces memory loss Reset count piece if it is the first start Otherwise we suggest to check if Battery is not discharged							
04/Feb/2017		14:58:03					

It indicates a memory loss of the count pieces or an initialization failure (that is the case in which it has never been put at zero).

If this screen appears, it is necessary to make zero in the count pieces. In case we are sure that the initialization has already been done and this message of error appear all the same, it is better to contact the assistance (it could be simply the inside battery discharged).

GOOD 0		BAD 0		▲		▼	
Reset?		Yes					
Last Reset:							
08:18		05/07/2016					
04/Feb/2017		14:58:03					

In the last line there is the indication of the date and of the hour of the last zero made (in case you have never put it at zero date and hour will contain casual values). The counter pieces rest procedure

allows to reset all the count of the first screen, and also to reset the memory of the last outcome (view the next screen description).

To put the equipment at zero it is necessary to move on “**Yes**” (that is similar to a parameter of modification, you will see it flashing) and you have to keep pushed the button **[+]** for about 3 seconds. Once you have put the equipment at zero, the tester will return automatically at the screen that shows the counts.

As already said, pushing the button **[E]** when you are on a result, it is possible to pass to a more detailed view of the result itself.

The first display that collects in a detailed way the parameters characteristic of the stored result of the test is the following one:

GOOD 99		BAD 86					
Test Num: 1				08:27:14 05/Feb/2017			
Result: Good				Prg: 001			
Leak test terminated in TEST							
Missing 0,00 [s]							
04/Feb/2017		14:58:03					
							

“**Test Num:**” is the number of the result, the same one that identifies this result in the first display.

On the same line of the number outcome there are the hour and date in which the test is finished: this are parameter logically univocal for each test (it is impossible that two test have the same hour and date, unless than a non-correct setting of the clock: pay attention at the solar and legal hour change).

“**Outcome:**” it represents the result of the test, and it is the same content in the first display. In brackets it can be add a further information that identifies in a more detailed way the kind of result (in the example the waste is due to the filling pressure that has gone below the tolerance). The messages are totally similar to those ones already explained in the chapter Frontal Panel. Here, a list with a description and the general outcome to which the particular belong:

- “**(Reco)**”, in the case of good, it indicates that it is a good with reservation, if the relative optional Third Result is active (view chapter of the optional one)
- “**(FL Lo)**”, in case of bad, it indicates a bad outcome due to flow under the threshold, if the relative optional Third Result is active (view chapter of the optional one)
- “**(FL Hi)**”, in case of bad, it indicates a bad outcome due to flow over the threshold, if the relative optional Third Result is active (view chapter of the optional one)
- “**(P% Lo)**”, in case of bad, it indicates a bad outcome due to the not attainment in the filling phase, or the fall in the settling phase, of the filling pressure less the tolerance (view chapter Test Cycle, and chapters Test Menu and Setup Menu for the relative parameter)

- “**(P% Hi)**”, in case of bad, it indicates a bad outcome due to the passing in the filling phase, or the passing in the settling phase, of the filling pressure more the tolerance (view chapter Test Cycle, and chapters Test Menu and Setup Menu for the relative parameter)
- “**(No Lev)**”, in case of bad, it indicates a bad outcome due to the not attainment in the filling phase, or the fall in the settling phase, of the filling pressure (view chapter Test Cycle, and chapters Test Menu and Setup Menu for the relative parameter)
- “**(BadRef)**”, in case of bad, it indicates a bad outcome due to the reference piece, that is the piece under test is better rather than the reference piece (outcome possible only for differential leak tester equipment Tx96x) (view chapter Test Cycle)
- “**(BadBell)**”, in case of bad, it indicates a bad outcome due to the bell, that is the piece under test have done a good outcome, but the bell check has changed the outcome in bad in the bell test phase (outcome possible only for bell leak tester equipment Tx98x) (view chapter Test Cycle)
- “**(Pover)**”, in case of bad, it indicates a bad outcome due to the attainment of the pressure full scale of the equipment
- “**(Flover)**”, in case of bad, it indicates a bad outcome due to the attainment of the flow full scale of the equipment (only for flow test equipment Tx7xx)

“**Prg:**”: represent the program number of the test of this outcome.

“**Test Decay finished in**” it is followed by the name of the phase in which the test is ending (the phase name in this case, like during the test cycle, it is ever in English language, independently of the language setting). The first word corresponds at the kind of test (view Test Menu) stored for this outcome.

“**when it lacks 0.00[s].**” take information about the moment in which the test is finished: being the test time in a backwards way, in this line it is showed the time that miss at the end of the phase (showed in the up line) in which the test is finished. Obviously, this time change from a test to another, and in the example, it is 6,14 seconds.

GOOD 99		BAD 86		▲	▼
Test Num: 1		08:27:14 05/Feb/2017			
Saved pressure		1000.3 [mbar]			
Saved VOUT		-0.180 [mbar]			
Saved VOUTAUX1		0 [mbar]			
04/Feb/2017	14:58:03				

The first line is the same of the first screen: it shows the outcome number associated at the test and the date and hour of it. Also, in this screen it is possible to scroll the outcome by changing the number that correspond to the outcome.

“**Stored Pressure:**” it is the same value contained in the first display.

“**Stored VOUT:**” it is the same value contained in the first display.

“Stored VOUTAUX1:” represent the test VOUTAUX1 stored. Decimal point and unit measure depend by the equipment and the activated optional.

According as the phase in which the test is finished, some of this value can be at zero, and their unit measure can be not correct. This show that this value is not be saved by the equipment because the test is finished before than their measure can be starting. For example, for an absolute decay leak tester equipment, the VOUT will be ever null for all the test finished before the test phase

Setup Menu

Setup Menu is the menu that contains the appropriate setting parameter of the specific equipment.

To enter in Setup Menu, you just press [SET] button on frontal panel of equipment. To exit from this Menu, press twice [SET] button to return in waiting status, press twice [TST] to exit from Setup Menu and accede directly to Test Menu.

Here below the menu screens are visualized in order of appearance into the display (from the first one to the last one) and its parameters are described: the chapter refers to the standard menu, that is without any optional activated. Some optional activate other sub menu, activate other parameters, modify the meaning of some standard parameters. All these modifications will be discussed in the chapter concerning the optional itself. In case that a parameter cannot be modified or selected, at the place of its value it will appear dashes "----".

First screen of SET Menu:

INSTRUMENT SETTINGS		▲	▼
Date setting:	05/02/2017	[----]	
Time setting:	08:49:35	[----]	
Keyboard brightness:	100	[%]	
Display minimum brightness:	100	[%]	
05/Feb/2017	08:49:35		

“**Date setting:**” it allows to regulate the date, format dd/mm/yyyy.

“**Time setting:**” it allows to regulate the hour, format hh/mm/ss (seconds cannot be changed).

“**Keyboard brightness:**” the value in percentage of the brightness of touch keyboard. During this parameter setting, the keyboard has and keep this value, in order to allow the operator to see the effect of this regulation.

“**Display minimum brightness:**” the value in percentage of the brightness in standby of LCD display. We talk about standby because the display has a fixed brightness of work when there is a result, when there is a start or a pressure of a button, while after 60 seconds after the pressure of a button, the brightness decreases till the work level. During this parameter setting, display has immediately the brightness of standby, not the maximum one, in order to allow the operator to see the effect of this setting.

Second screen of Menu SET:

INSTRUMENT SETTINGS		▲	▼
Serial address:	01	[----]	
Kind of Start:	From Panel	[----]	
Language:	English	[----]	
Graph visualization:	ON	[----]	
05/Feb/2017	08:49:35		  

“Serial address”: this parameter represents the ID of the equipment during serial/USB communication.

“Kind of start”: it allows to give the START to a test in different ways. According to the parameter value, modes are the following ones (see the chapter *Electric Signal* for further details):

- **“Panel excluded”:** excluded panel: START of the current program can be given with the activation of the input (pin 12) of the connector AUX1 combined to this mode. In this mode the mechanical button on frontal panel is not able to give the START.
- **“From Panel”:** the start of the current program can be given both with the activation of the correspondent input on the connector AUX1 (pin 12), and with the pressure of the mechanical button on frontal panel (see the chapter Frontal Panel).
- **“4 Start”:** the four inputs of the connector AUX1 combined with this mode (pin 4, 8, 9, 10), they allow with their activation to make the four associated programs to start (“001”, “002”, “003”, “004”). Keeping not activated the inputs of “4 Start” it is possible activating the input combined with the connector AUX1 (pin 12), to make the current program to start (that can be anyone of the 300 programs). In this mode, mechanical button on frontal panel is not able to give the START.
- **“BCD+STROBE”:** the four inputs of the connector AUX1 combined with this mode (pin 4, 8, 9, 10) allow by activating them to select in a binary logic BCD the desired program (the program “000” is excluded, that is the equipment is not sensible to the combination BCD 0000, then programs from “001” to “015” can be activated by this pin). When the input corresponding to the START is activated (pin 12 connector AUX1), this signal is interpreted as Strobe, and the program currently selected will start. Keeping BCD input not activated, it is possible activating the input combined with the connector AUX1 (pin 12), to make the current program to start (that can be anyone of the 300 programs). In this mode, mechanical button on frontal panel is not able to give the START.
- **“BCD+200ms”:** the four inputs of the connector AUX1 combined with this mode (pin 4, 8, 9, 10) allow activating them to select according to a binary logic BCD the desired program (the program “000” is excluded, that is the equipment is not sensible to the combination BCD 0000, then programs from “001” to “015” can be activated by this pin). After 200ms from the activation of the first input, the program currently selected will start. Keeping the inputs BCD not activated, it is possible activating the input combined with the connector AUX1 (pin 12), to

make the current program to start (that can be anyone of the 300 programs). In this mode, mechanical button on frontal panel is not able to give the START.

In any case, the pressure of the button [-] or the activation of the ABORT input (pin 13) of the connector AUX1, abort the running test cycle. Serial protocol functions are not affected by this parameter.

Besides, in case of start by the pressure of the button, it is necessary to be out from all the menu and that there is not any running test in order to give the start. In case of start from the connector AUX, the strategy to be adopted will be the same.

“Language:” this parameter allows to set the language of the equipment. Six languages can be selected, **“Italiano”**, **“English”**, **“Español”**, **“Français”**, **“Deutsch”**, **“Portugues”**. All the menu of the equipment visualizes messages using the language currently selected, unless the main menu and the menu of the running test, that are always in English.

“Graph visualization:”: this parameter allows to activate, if possible, the graphic display on cartesian axes of the typical measures of the test relating to the time. The possibilities are those ones:

- **“OFF”**: graphic not activated
- **“ON”**: graphic activated

In case of graphic of two traces in the same time, that is graphics both of the primary measure and of the secondary one, this parameter if it is different than **“OFF”** is referred to the secondary measure.

Third screen of SET Menu:

TIMES SETTINGS		▲	▼
Label signal duration:	1.0	[s]	
Acoustic signal duration:	0.5	[s]	
Minimum filling time:	1.0	[s]	
05/Feb/2017	08:49:35		  

“Label signal duration:” this parameter, if it is different from zero, identifies the time (in seconds) for which at the end of the test the output of the result combined with, activates, and the output concerning the running test remains active for all this time and the discharge time. In case this parameter is set at **“0.0”**, then the output of end of test disables at the end of the discharge time, and the output corresponding to the result will activate, and it remains active till the starting of a new test.

What we have described is valid only if the parameters **“End of cycle signal:”**, **“Bad management:”** and **“Abort management:”**, are all set on **“Standard”** (standard). Otherwise, the

signal behaviour of AUX1 connector is different (see here below the description of these three parameters), and this parameter is ignored.

“Acoustic signal duration:” this parameter sets the duration of the acoustic signal, in seconds. If it is set at zero, the acoustic signal is simply not activated (unless the cases in which the acoustic signal has a duration fixed at firmware and not depending on this parameter).

“Minimum filling time”: this parameter allows to set a minimum filling time. If the equipment fills the piece under test before set time in this parameter, so it will give bad result.

Fourth screen of SET menu:

MEASUREMENT SETTINGS		▲	▼
Pressure filter mean number:	5	[----]	
VOUT filter mean number:	5	[----]	
Direct pression dec.fac.vis:	2	[----]	
05/Feb/2017	08:49:35		  

“Pressure filter mean number:” this parameter indicates on how many values the direct has to be mediate Much greater it is the number of values with which you have to make an average and much greater it will be the stabilization of the measure but the speed of variation of the same one will be lower, vice versa, much lower it is the number of values with which you have to make an average, less it will be filtered, resulting in this way less stable. Generally, for highest speed tests, you have to reduce it at minimum, for long tests instead you have to raise it.

“VOUT filter mean number:” it has the same function as the previous one, but it acts on the second dimension that the equipment measures.

Please note: In Tx99x instruments this parameter cannot be set, and is bound to be equal to the relative direct pressure.

“Direct pression dec.fac.vis.:” (Direct pression decimal factor visualization): in order to make more readable and clearer the direct pressure measure, considering the high resolution of the measurement inside the equipment necessary to calculations, with this parameter we can reduce the number of decimal points or of less significant digits of the value displayed, without reducing or compromise in anyway the goodness of total measures.

We underline again, that this parameter only affects the display, but in any way the resolution of measure of the equipment.

Fifth screen of SET menu:

TEST SETTINGS		▲	▼
Tolerance on the opp. decay:	100	[%]	
At a bad outcome:	Stop the test	[----]	
Equalization during settling:	----	[%]	
05/Feb/2017	08:49:35		  

“Tolerance on the opp. decay:” (tolerance on the opposite decay): this parameter defines a tolerance limit (calculated on the maximum admitted decay set in the chapter *Test Menu* **“Maximum leak:”**, on spurious phenomena: in fact we talk about of inverted decay because this limit is activated when the decay instead of growing (decreasing) decreasing (increasing). We assume to be simple that the test is in pressure, and not in depression (in this case discussion is inverted). In absolute decay leak testers, we can talk about a positive decay, because the pressure inside the piece when test is running instead of decreasing it increases. This parameter is useful to define a limit under which the piece can yet be considered as good one: above this limit, the equipment gives a result of positive decay **“POS LEAK!”** (considered as an abort). If we take the for example a Tx99x equipment under pressure: if admitted pressure is 10 mbar and the parameter in the description has a value of 50%, then in case that the pressure inside the pieces would increase not over +5 mbar (10 mbar x 50%) the piece is considered good, while if it will increase moreover, the equipment will give as result a positive decay. Generally, inverted decay should not happen, because it is an indicator of wrong test parameter, temperature problems, dilation of the piece in test, etc...

“At a bad outcome:”: this parameter indicates the way the equipment behaves in case of overcoming of the limit of maximum decay admitted in leak test phase. Particularly, according to the selected mode:

- **“Continue the test”**: the limit passed, the equipment continues the test till setting test time is finished, and it continues to measure and to visualize the decay.
- **“Stop the test”**: once you overcome the limit, the equipment stops the test even if timing test is not finished, and it visualizes as a decay, the maximum admitted decay set (**“Maximum leak:”**).
- **“Stop and calculate”**: once you overcome the limit, the equipment stops the test even if timing test is not finished, and it visualizes as a decay a predictor number about what it would be the decay measured at the end of timing test.

The equipment is sensible to this parameter only if the type of measure of the decay in test menu (see chapter *Test Menu*) is pressure. (**“Pressure”**): in all the other cases, the equipment will behave as if this parameter would be set on **“Continue the test”**.

“Equalization during settling:” this parameter expresses, in percentage, a time in the settling phase during which at the beginning of the settling phase certain actions are performed.

During this time the equalization valve continues to keep the reference and test side in communication. This function allows to improve an unbalance phenomenon made by the two measuring branches.

As said, this parameter is in percentage and it is linked to the parameter "Settling phase duration:" of the Test Menu in order to transform it into a time: the valve remains open for a time equal to "Settling phase duration:" x "Equalisation in settling:" /100: if for example "Settling phase duration:" is "10.0" s and "Equalisation in settling:" is "30" %, then the valve will remain open for the first 3 s of the settling phase.

Sixth screen of SET Menu:

COM PORTS SETTINGS		▲	▼
SER0/USB modality:	Nothing	[-----]	
SER0/USB Baud Rate:	115200	[bps]	
SER1 modality:	Barcode	[-----]	
SER1 Baud Rate:	9600	[bps]	
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“SER0/USB modality:” COM1 port of the equipment different functions can be possible according to the value of this parameter:

- **“Nothing”**: any type of communication and of protocol is activated on this port.
- **“ForTest Prot.”** (ForTest Protocol): in this mode is active the protocol of the type "you ask, I answer", between an RS232 terminal and the instrument. This communication is present on the RS232 interface (COM1 connector), on the RS485 interface and on the USB connector.
- **“ForTest Prot. TTY”**: like the case **“ForTest Prot.”**, but in the hardware side, the communication is a ring current. TTY interface is available on COM2 connector.
- **“Trace end of test”**: the equipment at the end of the test sends a trace (a series of characters) with specific parameters of the same one. This modality is not present on TTY interface but on the other hardware interface associated to SER0 communication.
- **“Modbus RTU”**: in this modality Modbus protocol RTU is activated. This type of communication is not present on TTY interface but is present on the other hardware interface associated to SER0 communication.

Please note: SER0 communication is present on all hardware resources described on chapter **“Electric components Back panel”** but **NOT at the same time. This means that is not possible, for example, using at the same time RS485 interface and USB but only one by one.**

Please note 2: For further information on protocols and transmitted data, see chapter of Serial Communication.

“Baud Rate SER0/USB”: with this parameter is possible to select the speed in bsp (bit/second) of communication of SER0 serial. Possible values are **“9600”**, **“19200”**, **“38400”**, **“57600”**, **“115200”**. Speed is in common with all the available hardware interfaces.

“SER1 Modality”: as for **“SER0/USB Modality”** parameter, but physically the communication is only on COM1 port, SER1 serial. It is possible to choose following functionalities:

- **“Nothing”:** Any type of communication or protocol is activated on port.
- **“Barcode”:** On the port, by a barcode reader, it is possible to read barcodes.
- **“Lola EPL”:** A label can be printed on the door using a LolaLP2 printer with a peeler. Use the EPL printing language.
- **“LP2824 EPL”:** as in the case of **“Lola EPL”**, but for Zebra LP2824 printer.
- **“ZD410 EPL”:** as in the case of **“Lola EPL”**, but for Zebra ZD410 printer.
- **“Zebra ZPL”:** on the door, through the use of a Zebra printer using the ZPL language, it is possible to print a label.

Please note: Choosing a printer, remains activated the communication through barcode.

“Baud Rate SER1:” See analogue parameter on **Baud Rate SER0/USB:** the meaning is the same.

Seventh screen of the SET menu:

WIRELESS SETTINGS		▲	▼
Connection type:		Off	[-----]
Connection status:		Disconnected	
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“Connection type:” This parameter allows you to set the type of wireless connection to be activated on the instrument. This parameter can take on these values:

- **“Off”:** the wireless card is off.
- **“BT2.0”:** activates Bluetooth 2.0. The device will be visible with the name "FORTEST-SNXXXX" where "XXXX" is the serial number of the instrument.
- **“BLE”:** activates Bluetooth Low Energy. The device will be visible with the name "FORTEST-SNXXXX-T8YYY" where "XXXX" is the serial number of the instrument and "YYY" is the model.

“Connection status:” Indicates the status of the instrument: "Disconnected" if no device is connected to the instrument, "Connected" if an external device is connected to the instrument. This parameter cannot be changed but only displays the connection status.

Eighth screen of SET menu:

PASSWORD SETTINGS: UNBLOCKED		▲	▼
Lock level:		1: Parameters	
What do you want to do?		Block?	
Enter password:		*****	
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The equipment contains four passwords of 10 alphanumeric characters per each one (code ASCII from 33 to 126 including extremes):

- One is selected by the Customer and it can be set, at the birth of the equipment is set on **"0000000000"** (PWCLIENTE).
- One is for recovery in case the Customer loose his password, supplied by the company (PWMAGIC1).
- One that allows temporarily to unblock the equipment (PWTEMP) (the equipment passes at the state **"TEMPunblocked"**): temporarily it means that if the equipment is turned off it will come back to **"Blocked"** state.

The default password at first power on is "0000000000", so if you change it, enter it in the string "Enter password".

The title of the screen varies depending on the status of the machine:

- **"Unblocked"**: means that the instrument parameters can be changed. This condition is remembered at the turn off: if the equipment turns off when it is in this status, and after it is turned on, you will find it again in this condition.
- **"Blocked"**: means that you are not allowed to change the instrument parameters. This condition is remembered when the equipment turned off: if the equipment is turned off when it is in this condition, and after it is turned on, you find it again in this condition.
- **"TEMPunblocked"**: means that the instrument parameters can be changed. This condition is not remembered when you turned off the equipment, if the equipment is turned off when it is in this condition, and then it is turned on, you will find it in the state **"Blocked"**.

To pass from a state to another one it is necessary to insert a correct password.

From the state **"Unblocked"** you can pass to the state **"Blocked"**.

From the state **"Blocked"** you can pass at the state **"Unblocked"** inserting the passwords PWCLIENTE (set by the Customer) and PWMAGIC1 (from the factory, contact the assistance to

know this password). From the state **“Blocked”** you can also pass at the state **“TEMPunblocked”** inserting the password PWTEMP.

From the state **“TEMPunblocked”** you can also pass at the state **“Unblocked”** inserting passwords PWCLIENTE and PWMAGIC1.

“Lock level:”: With this parameter you can set what to lock with the password. When the machine is unlocked the padlock icon is green . Possible blocking levels are:

- **“1: Parameters”**: With level 1, changes to all parameters in the TST and SET menus are blocked. The padlock icon will turn yellow .
- **“2: Param.+Prog.”**: With level 2, changes to all parameters in the TST and SET menus and program changes are blocked. The padlock icon will turn orange .
- **“3: Par+Prg+TST”**: level 3 blocks the changes to all the parameters in the TST and SET menus, the program change and the selection of the TST key. The padlock icon will turn red .

“What do you want to do?”: with this parameter you can tell to the equipment which kind of action connected to the parameter block you want to do:

- **“Unblock?”**: the operator wants to disable the parameters block. As we said the unblock can be definitive or temporary: which one of the two will be put in action, depends from the password that will be insert.
- **“Block?”**: the operator wants to activate the parameter block.
- **“Modify PWD?”**: the operator wants to modify the PWCLIENTE.

If the equipment is in the **“Blocked”** state, the only action allowed is **“Unblock?”**.

If the equipment is in the **“Unblocked”** state, the operator can **“Block?”** or **“Modify PW?”**.

If the equipment is in the **“TEMPunblocked”** state, the only action allowed is **“Unblock?”**.

“Enter password:”: in this parameter it is necessary to insert the password. The first message that will appear to the operator is **“Press [E]”**: by pressing the key you can start to set the password using the alphanumeric keypad. At this point you can enter the password, like any other string. Pressing the green "E" key, different messages could appear:

- **“WRONG PWD”** if the password put in does not correspond to any of the correct ones.
- **“CORRECT PWD”** if the password put in is correct, and it is different from the temporary one.
- **“PWD TEMP OK”** if the password put in is correct, and it coincides with the temporary one

“Enter new password:”: this parameter is active and visible only if the selected action is **“Modify PWD?”**. To enter this section, first of all it is necessary that you have insert a correct password previously (section **“Enter password:”**), and different from the temporary password. New password insertion works in the same way as the password insertion at the section **“Enter password:”**. Pressing the green **“E”** key, the message will appear is **“PWD SAVED ”**, to show the actual saving of the new password.

Ninth screen of SET menu:

VARIOUS SETTINGS		▲	▼
Auto-Zero pressure every:	0	[min]	
Result leds max. brightness:	100	[%]	
Enable messages end test:	Yes	[-----]	
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“Auto-Zero pressure every:”: it allows to set a cyclic time at the end of which the equipment makes automatically the auto zero procedure. If this parameter is set to zero, we mean that the automatic auto zero proceeding has been excluded, while it remains obviously possible to make the auto zero by the button (see chapter *Frontal Panel*). For further information on the same proceeding, please refer to the chapter *Frontal Panel*.

“Result leds max. brightness”: it allows to set the brightness intensity of the indication led of test result.

“Enable message end test:”: it allows to enable messages end test.

Tenth screen of SET menu:

PLC SETTINGS		▲	▼
End of cycle signal:	Standard	[----]	
Bad management:	Standard	[----]	
Abort management:	Standard	[----]	
Output kind:	----	[----]	
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This menu contains some parameters that allows to modify the behaviour of output and input signals placed on AUX1 connector, but they do not influence on the input that the operator can give from the frontal panel (by the buttons) or outputs on the frontal panel (visualized on the display or on the led). They influence the behaviour of the equipment (for example on the automatic activation or by external activation of the solenoid discharge), and they are thought typically for an exclusive use of

the equipment with a PLC (or another automation or with an appropriate keyboard and indicator light).

More specifically, the values taken by these three parameters, allow a different strategy of management of the good pieces but mostly of the waste ones and of the abort ones (for example if you disable the automatic discharge keeping the testing piece under pressure, you can search the leak with soap or foam methods), a different activation of the signal that indicates running/end test (and consequently the different moments in which you can give the new start). Here below we describe in words how these parameters working, and we send to the appropriate chapter *Timing Diagram* for a better comprehension, and also to the chapter concerning the electric signals.

NEW START: according to the value of the following parameters, it changes the moment in which the new start is allowed:

- if **“Bad management:”** / **“Abort management:”** and **“End of cycle signal:”** has a value of **“Standard”**, and **“Label signal duration:”** is null, a new start can be given once that the **“Discharge phase duration:”** time is finished (in the case that the discharge time is null, is like than the discharge is not active). See chapter *Test Menu* for the case of program linked.
- if **“Bad management:”** / **“Abort management:”** and **“End of cycle signal:”** has a value of **“Standard”**, but **“Label signal duration:”** it is not null, a new start can be given once that both the **“Discharge phase duration:”** time and the **“Label signal duration:”** time are finished (in the case that the discharge time is null, is like than the discharge is not active; we remind that the **“Label signal duration:”** time start at the end of the test and it is the time for which the outcome output remain active). See chapter *Test Menu* for the case of program linked.
- if **“Bad management:”** / **“Abort management:”** has a value **“All Automatic”** and **“End of cycle signal:”** has a value **“Busy”** or **“Ready”** or **“Pulse”**, then a new start can be given after the end of the discharge.
- if **“Bad management:”** has a value **“Validation”** and **“End of cycle signal:”** has a value **“Busy”** or **“Ready”**, then a new start can be given after the end of the discharge.
- if **“Bad management:”** has a value of **“Validation”** and **“End of cycle signal:”** has a value of **“Pulse”**, then a new start can be given after the cycle test end impulse.
- if **“Bad management:”** / **“Abort management:”** has a value of **“Remain Full”** and **“End of cycle signal:”** has a value of **“Busy”** or **“Ready”**, then a new start can be given after the discharge end.
- if **“Bad management:”** / **“Abort management:”** has a value of **“Remain Full”** and **“End of cycle signal:”** has a value of **“Pulse”**, then a new start can be given after the cycle test end impulse.

“End of cycle signal:”: by this parameter it is possible to change the function of the end cycle signal (PIN 5 connector AUX1) according to the value that has been set:

- **“Standard”**: the signal is activated when you give the START to the test, and deactivated when both the **“Label signal duration:”** time, which start at the beginning of the discharge, and the **“Discharge phase duration:”** time are finished. See chapter *Test Menu* for the case of program linked.
- **“Busy”**: the signal is activated when you give the START to the test, and deactivated at the end of the discharge when there is the end of the test.
- **“Ready”**: exactly dual of the **“Busy”** case.
- **“Pulse”**: impulse at the end of the test: as in case of **“Ready”**, but instead of activating the signal, at the discharge end and when the test is finished, after a pause of 500ms it is generated a pulse of the duration of 200ms.

“Bad management:”: by this parameter it is possible to modify the working of waste signal (PIN 6 connector AUX1), of the signal of good and of the input of abort (and indirectly of the signal of test end) placed on the connector AUX1 according to the value that has been set:

- **“Standard”**: the waste or good signal is activated at the beginning of the discharge phase, and it is deactivated at the end of the parameter **“Label signal duration:”** or at the start of a new test; if the parameter **“Label signal duration:”** is placed at **“0.0”** s, the waste signal or the good signal remains activated till a new start. The discharge will start automatically. See chapter *Test Menu* for the case of program linked.
- **“All Automatic”**: at the end of the test phase the waste or good signal is activated and the discharge will begin automatically; at the end of the discharge also the test ends (and there is the activation of the signal of test end according to the parameter **“End of cycle signal:”**, and consequently the possibility to give a new start): the waste or good signal is deactivated at the start of a new test or it can be deactivated with an ABORT impulse.
- **“Validation”**: if the result of the test is good, the behaviour is the same as in the case **“All Automatic”**. If the result is waste, at the end of testing phase the waste signal is activated and it begins the discharge automatically; only the reception of an abort signal (to receive all the way at the end of the discharge) will give an end to the test (and then the activation of the signal of test ending according to the parameter **“End of cycle signal:”**, and consequently the possibility to give a new start) and the deactivation of waste signal (if the parameter **“End of cycle signal:”** has a value of **“Pulse”**, the waste signal is deactivated at the end of 500ms of delay, that is together with the rising edge of the signal of test ending).
- **“Remain Full”**: if the result is good, the behaviour will be the same as in the case of **“Auto”**. If the result is waste, at the end of testing phase the waste signal will be activated; the piece remains filled till the abort signal will arrive: at this point the discharge will begin, and at the

end of the same one there will be the test ending (and then the activation of the signal of test ending according to the parameter **“End of cycle signal:”**, and consequently the possibility to give a new start) and the waste signal is deactivated (if the parameter **“End of cycle signal:”** has a value of **“Pulse”**, the waste signal is deactivated at the end of 500ms of delay, that is together with the rising edge of the signal of test ending).

“Abort management:”: by this parameter it is possible to modify the working of abort input (and indirectly of ending test signal) placed on pin 13 of the connector AUX1 according to the value that has been set:

- **“Standard”**: if during a test (in any case before the beginning of the discharge) the equipment receives an abort signal, this one has the same effect as the pressure of the button [-]: the test will stop and the discharge phase starts.
- **“All Automatic”**: if during a test (in any case before the beginning of the discharge) the equipment receives an abort signal, the test will stop and the discharge starts: at the end of the discharge there is the end of the test (and then the activation of the end testing signal according to the parameter **“End of cycle signal:”**, and consequently the possibility to give a new start).
- **“Remain Full”**: if during a test (in any case before the beginning of the discharge) the equipment receives an abort signal, the test stops: the piece remains filled till it does not receive a new abort impulse (at least 500ms from one to the other). At this point the discharge starts, and at the end of the discharge there is the test ending (and then the activation of the signal of test ending according to the parameter **“End of cycle signal:”**, and consequently the possibility to give a new start).

Please note: not all the combinations of the three parameters are allowed, both for functional reason and logical ones. In particular these combinations are allowed (that are in any cases forced as choice level from the equipment):

- **“Bad management:” = “Standard”, “Abort management:” = “Standard”, “End of cycle signal:” = “Standard”**.
- **“Bad management:” = every value except “Standard”, “Abort management:” = every value except “Standard”, “End of cycle signal:” = every value except “Standard”**.

The combination **“Bad management:”= “Standard”, “Abort management:”= “Standard”, “End of cycle signal:”= “Standard”**, is that one that has a behaviour as compatible as possible with the use of the frontal panel (indeed the difference is in the possibility to give a new start: in the button case, it is not necessary to wait the end of the discharge time and of the label time). For the other combinations, it is better to set the parameter **“Kind of Start:”** on a different value of **“From Panel”**, in order to avoid management conflict.

Eleventh screen of SET menu:

TEST SETTINGS		▲	▼
Control at % settling time:	0	[%]	
Recharge at % filling time:	0	[%]	
Settling's attempts:	0	[----]	
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“Control at % settling time:” This parameter indicates after what adjustment time percentage the filling pressure control starts. In the event the pressure drops below the filling pressure tolerance, the equipment shall refill the piece.

“Recharge at % filling time:” This parameter indicates the filling time in the event a refill is carried out. It is indicated in the filling time percentage.

“Settling's attempts:” In the event that, at the end of the adjustment phase, the pressure drops below the threshold established by: **“Filling pressure:”** +/- **“pressure tolerance:”** And this parameter is other than zero, the equipment will restart a new filling phase. This can be repeated up to n times, where n is the value of this parameter: Once this number n has ran out, the outcome of the test shall be negative.

Twelfth screen of SET menu:

PRINTING SETTINGS (1/2)		▲	▼
Format labels:	60x25	[----]	
Print also linked:	No	[----]	
Print also bad:	No	[----]	
Print also abort:	No	[----]	
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“Format labels:” parameter that allows to choose format label to print. The possible values are 60x25 (labels with max 5 rows), 60x60 (labels with max 15 rows) and Personalized (labels with max 16 rows).

“Print also linked:” parameter that allows to set up or not a print for each test done on the global interlocked cycle.

“Print also bad:” parameter that allows to set up if print the label relative to a finished test with bad result.

“Print also abort:” parameters that allows to set up if print in case of abort.

PRINTING SETTINGS (2/2)		▲	▼
Labels to print:	1	[----]	
ZPL print resolution:	203 dpi	[----]	
Reset progressive:	No	[----]	
Progressive:	0	[----]	
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To access this screen, select the "**Print also abort:**" parameter and then press the [E] key or the [▼] key.

"Labels to print:" number of labels: parameter that allows to set up how many labels to print at the end of the test about single test.

"ZPL print resolution:" allows you to adjust the print resolution for printers that use the ZPL language. It can take 203, 300 and 305 dpi as values.

"Reset progressive:" allows you to reset the value of the "**Progressive:**" parameter. Setting it to "**Yes**" resets the progressive value.

"Progressive:" Invariant item. Shows the value of the stored progressive. This value increases by one for each label printed. Ranges from 0 to 999. When 999 is exceeded, it starts again from 0.

Thirteenth screen of SET menu:

TICKET SETTINGS (1/4)		▲	▼
Row 1:	Name and num. prog	[----]	
Row 2:	Date and time	[----]	
Row 3:	Decay in test	[----]	
Row 4:	Result	[----]	
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This menu can be composed of up to four pages depending on the format of the chosen label (2 page for the format 60x25, 4 for 60x60 e Personalized). Each parameter of these pages corresponds at a row that will be printed on the label and it can take over one of the following 23 values:

- **"None"**: it avoids row insertion.
- **"Empty line"**: it prints an empty row.
- **"Name e num. prog."** (name and number program): it prints a row with the program name and number.
- **"Program name"**: it prints a row with the name of the program.
- **"Program number"**: it prints a row with the program number.
- **"Date and time"**: it prints a row with the date and time of the test performed.

- **“Barcode”**: it prints a row with the barcode read for the test.
- **“Test type”**: it prints a row with the type of test performed.
- **“Fill. pressure”** (filling pressure): it prints a row with the value of the filling pressure set for the test.
- **“Decay” / “Flows Min-Max”**: prints a line with the value of the drop set for the test if the instrument is leak-proof, while for flow-proof instruments it prints the minimum and maximum flow rates.
- **“Filling type”**: it prints a row with the type of fill set for the test.
- **“Settling time”**: it prints a row with the settling time set for the test.
- **“Test time”**: it prints a row with the time set for the test.
- **“Last ph. perf.”** (last phase performed): it prints a row with the last step performed for the test.
- **“T. last ph. per.”** (time last phase performed): it prints a row with the time of the last step performed for the test.
- **“Press. in test”** (pression in test): it prints a row with the value of the pressure relative to the test.
- **“Decay in test” / “Flow in test”**: prints a line with the value of the drop relative to the test if the instrument is leak-proof while for flow-proof instruments it prints the flow rate in test.
- **“Result”**: it prints a row with the result of the test.
- **“Company”**: it prints a row with the set company name.
- **“Operator”**: it prints a row with the set name of the operator who performed the test.
- **“Serial number”**: it prints a row with the serial number of the instrument that performed the test.
- **“Lot code”**: prints a line with the inserted lot code.
- **“Progressive”**: prints a line with the progressive value.

Please note: The values **"Nothing"** and **"Empty line"** can be repeated several times while the other values can appear only once in the label.

Fourteenth screen of SET menu:

BARCODE SETTINGS		▲	▼
Barcode reading request:	No	[----]	
Autostart from barcode:	No	[----]	
Code type:	Ware	[----]	
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“Barcode reading request:” request to read a barcode to start each test: parameter that set to **"Yes"** requires the reading of a barcode before starting each test.

“Autostart from barcode:” start the test by reading a barcode: parameter that allows you to start a test automatically as soon as a barcode is read.

“Code type:” it specifies the type of code: parameter that allows to specify if the barcode read is relative to a single ware (piece) or to a lot.

Fifteenth screen of SET menu:

BARCODE SETTINGS		▲	▼
Program start index:	1	[----]	
Length:	1	[----]	
Barcode program:	No	[----]	
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“Program start index:”: starting position of the program in the barcode: parameter that indicates the starting position of the program in the barcode read; it can take on a value between 1 and 40.

“Length:”: size of the program name: parameter that indicates the number of characters that make up the name of the program; it can take on a value between 1 and 16 (becomes less than 16 if the start index of the program is greater than 25).

“Barcode program:”: search a program by reading a barcode: parameter that if set to "Yes" allows the search and positioning on a program reading a barcode and extracting the program name from the barcode using the set program start index and length.

Sixteenth screen of SET menu:

COMPANY SETTINGS		▲	▼
Company:	FORTEST	[----]	
Operator:	MARIO ROSSI	[----]	
Lot code:	AX129345J	[----]	
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“Company:”: it allows you to set the company name (max 16 characters). The company name remains saved even when the instrument is restarted. When the parameter is being modified, it is possible to enter the name of the company by means of a barcode reading.

“Operator:” it allows you to set the operator name (max 16 characters). The name of the operator is lost when the instrument is restarted. When the parameter is being modified, it is possible to enter the name of the company by means of a barcode reading.

“Lot code:” allows you to set a lot code (max 16 characters). The batch code remains saved even when the instrument is restarted. When the parameter is being modified, it is possible to enter the batch code by means of a barcode reading.

Seventeenth screen of SET menu:

CALIBRATION OF THE INSTRUMENT		▲	▼
Enter the calibration menu?	Yes	[----]	
Last calibration date:		[----]	
Pressure:	25/02/2017 08:56	[----]	
VUOT:	25/02/2017 08:57	[----]	
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This last screen allows to accede to calibration menu of leak tester, for further details about it see chapter *Calibration Menu*.

Version Menu

The Version Menu is the menu that summarizes the particular characteristics of the equipment.

To enter to the Version Menu is enough to push twice **[SET]** button, present on frontal panel of equipment. To exit, push again **[SET]** button to come back to waiting status, press **[TST]** button directly to Test menu.

Here below the menu screens are visualized in order of appearance into the display (from the first one to the last one) and its parameters are described: the chapter refers to the standard menu, that is without any optional activated. Some optional activate other sub menu, activate other parameters, modify the meaning. In case that a parameter cannot be modified or selected, at the place of its value it will appear dashes "----".

First screen of Version Menu:

INSTRUMENT VERSION		▲	▼
Direct pressure:	6000.0	[mbar]	
VOUT channel:	6000.0	[mbar]	
Serial number:	6387	[----]	
03/Feb/2017	14:58:09		  

“Direct pressure:”: full scale in pressure of the equipment.

“VOUT channel:”: VOUT equipment full scale.

“Serial number:” Serial number: serial number of the equipment, each equipment has a unique serial number, belonging to any model

Second screen of Version Menu:

INSTRUMENT VERSION		▲	▼
Instrument code:	T8990-065000-530	[----]	
(optional)	080000048000	[----]	
Checksum:	1B16	[----]	
Checksum boot:	5CEF	[----]	
03/Feb/2017	14:58:09		  

“Instrument code:”: pneumatic optional: this parameter identifies the model of equipment.

“Optional:”: Equipment optional: this parameter identifies the optional activated in the equipment.

“**Checksum:**”: value of the checksum of the instrument's firmware. Parameter displayed in hexadecimal format.

“**Checksum Boot:**”: value of the instrument boot checksum. Parameter displayed in hexadecimal format.

Calibration Menu

Calibration Menu is the menu that sets the adjustment of pressure measurement of equipment. Moreover, allows to set calibration parameters of the equipment and of the working settings and the setting of the same one.



ATTENTION: The access to this menu can create measure errors, malfunctions caused by accidental incorrect setting, or settling modifications from operator. It is suggested to contact ForTest Italia srl before entering to this menu. After this access, always check with master pieces the correct functionality of equipment.

To entry to this Calibration menu it is necessary press once **[SET]** button on frontal panel of equipment, press twice **[▼]** button until arriving the following screen:

CALIBRATION OF THE INSTRUMENT		▲	▼
Enter the calibration menu?		Yes	[----]
Last calibration date:			[----]
Pressure:	25/02/2017 08:56		[----]
VUOT:	25/02/2017 08:57		[----]
05/Feb/2017	08:49:35		  

After a check of calibration with no-compliance result, you need to adjust the tester measure. This operation can be done by specialized staff nearby Fortest Italia's laboratory, or by customer.

We suggest to assign this operation to staff with adequate level of qualification and before to proceed, contact ForTest Italia Srl to ask for the correct procedure to adjust the measure.

Service Menu

The equipment has a mode that allows a test of the hardware, a restoration of initial settings, an updating of the equipment firmware. Seen the functionality of this menu, it is recommended not to use it unless it is clearly indicated by our assistance.



WARNING: The access to this menu can create measure errors, malfunctions caused by accidental incorrect setting, or settling modifications from operator. It is suggested to contact ForTest Italia srl before entering to this menu. After this access, always check with master pieces the correct functionality of equipment.

To accede to this mode, press twice **[SET]** button, then **[▼]** button twice. At this point you can reach this screen:

SERVICE					
Enter the service menu?				Yes	[----]
05/Feb/2017		08:49:35			

At this point press **[E]** button, “**Yes**” writing is selected, so keep pressing **[+]** button, screen will be updated:

Press [+] button for FLASH MODE					
Press [-] button for TEST MODE					
Press [SET] button for program STP					
Press [TST] button for program TST					
Press [E] button to exit					
05/Feb/2017		08:49:35			

As you can deduce also from the text, the pressure of the two different buttons allows different functions:

- **Button [+]:** it allows to make the equipment entering in flash mode. This mode is made in order to obtain an updating through serial or USB port of installed firmware on equipment. For more details, see following paragraph “FLASH MODE”.

- **Button [-]:** it allows to make the equipment entering in test mode. This mode allows to verify hardware functionality of the equipment. Before entering in this mode, we recommend to disconnect everything is connected on the pneumatic junctions of the equipment.
- **Button [SET]:** it initializes all the setup and calibration parameters (unless the calibration of the transducer/s of measure) to default values. NB: for default value we mean general values for the particular model of the equipment, and then not the specific ones of the particular application of the Customer. This proceeding is then surely to be made at the first start up of the equipment, but if the equipment had been already started and initialized, it is to avoid if not for cases advised from our assistance.
- **Button [TST]:** it initializes all the program parameters to default values. NB: for default value we mean general values for the particular model of the equipment, and then not the specific ones of the particular application of the Customer. This proceeding is surely to be made at the first start-up of the equipment, but if the equipment had been already started and initialized, it is to avoid if not for cases advised from our assistance.
- **Button [E]:** it allows to go out from the menu.

This procedure is therefore certainly to be carried out the first time the instrument is switched on, but if the instrument had already been switched on and initialised, it should be avoided except for cases recommended by our assistance.

FLASH MODE

After the button pressure, the equipment will ask in which way you want make flash mode and performing an updating:

[TST] for COM1 FLASH		▲	▼
[SET] for [USB] FLASH Press [E] button to exit			
05/Feb/2017	08:49:35		
			

It is possible to perform the updating in two different ways: through RS232 (pressing “TST” button) or through USB key (pressing “SET” button).

- **[TST] for COM1 FLASH:** it allows to update the equipment through FlashTerminal software (supplied by ForTest). Please contact ForTest for this type of updating and documentation about it.
- **[SET] for [USB] FLASH:** it allows to update the equipment with only USB key. It is necessary that on USB pen there is a directory with the name of equipment (for ex. T8990); inside this directory should be present a firmware supplied by ForTest. If the tester detect correct directory and firmware, will start automatically the procedure of updating.

- **Press [E]** button to exit: it allows to exit and restart the equipment.

During the update, display visualize the writing “Loading Firmware”; RED RGB led will turn on to indicate to not press other buttons during the updating. Two lines of blue led indicate the evolution of the updating. At the end of the procedure, leak test will automatically re-start.

The only entrance through button pressure in this mode, do not delete the memory of equipment: so if you do not connect anything to serial connector or anything is present on USB key, it is possible to turn off equipment and then turn it on.

Please note: Firmware updating is a very delicate procedure and can bring to a complete NO-WORKING of equipment. Be careful and let that ForTest will guide you in order to perform correctly the update of equipment.

Parameter Backup and Restore Menu from USB

Parameter Backup and Restore Menu from USB pen allows to import and export, through an external USB unit, program and set up parameters of equipment.

Therefore, it is possible to save TST program preset or SET (setting), then copy them on a new one equipment (if the two testers are similar in hardware and firmware), restore always on the same tester of preset or set up previously saved.

To enter in the menu, it is necessary:

- Check if USB pen has been inserted in the frontal panel of the unit and that the symbol  is not crossed out.
- Press symbol  from main menu (from [TST], [SET] and [CTP] menu is not recalled)
- Check that none test is underway.

From here on, USB unit in the drive will be used as destination for possible backup or as source for data.

Screen of menu:

BACKUP/RESTORE FROM USB PEN			
☒	Export TST from program	0	to 299
☐	Import TST from program	0	to 299
☐	Export SET	☐	Import SET
START		Waiting	   

Round boxes indicate that the operation will be done. To change the operation, touch the box. Multiple operations are not allowed, so only one box can be selected at a time.

Squared boxes are referred to present operations on the same line and indicate intermission on which operation will be done. To change these pauses, touch the box and use [+] and [-] buttons present on panel to change the value.

"**START**" button has the role to start the operation selected actually.

Progress bar with the writing "**Waiting**" has the purpose to show progression phase of the underway operation.

Possible operations are:

- **Export TST from program [n] to [N]** : copy on USB (present on drive) all set up parameters from program [n] to program [N], with $[n] \leq [N]$. Parameter [n] indicate the

program from which will start the exportation. Parameter [N] indicate the last program from which will be completed the exportation. Wider will be the program range, bigger will be the time necessary to do the selected operation.

- **Import TST from program [n] to [N]** : import from USB (present on drive) all parameters previously saved from program [n] to program [N] , with $[n] \leq [N]$. Parameter [n] indicate the program from which will start the importation. Parameter [N] indicate the last program from which will be completed the importation. In case is indicated a bigger range of intermission compared to that one saved on USB, the writing "*Err. file prg*" will compare in order to indicate that more parameters have been uploaded. Wider will be the program range, bigger will be the time necessary to do the selected operation.
- **Export SET**: copy on USB (present on drive) SET menu parameters. For this operation, additional parameters are not expected.
- **Import SET**: import from USB (present on drive) SET menu parameters previously copied. It is not possible to import parameters from a different equipment from which has been copied. The check-up will be done on serial number of testers. For this operation additional parameters are not expected.

Possible parameters already present on USB unit will be overwritten in case of exportation.

To exit from menu is necessary to press one of menu buttons [TST], [SET] or [CTP].

Back panel, pneumatic connections

This chapter gives basic information on the right installation of the equipment. We have to premise that these operations must be made by qualified staff.

- Keep the equipment far from heating source or from cold one to avoid phenomena of thermal drift, that would distort the running test. In fact, the pressure is directly proportional to the temperature, then, in testing phase, if we had a temperature change, we would have by consequence an increase (in case of heat) or a decrease (in case of cold) of the pressure (and therefore of the decay and flow too).
- Use compressed air at ambient temperature, filtered and without oil, to avoid fault or malfunctions at the pneumatic circuit. We recommend a two-stage filter of 25 micron and 1/100 micron before the pneumatic supply of the equipment.
- Pieces under test must be at ambient temperature to avoid false tests due to temperature increasing during the test.
- Use for the pneumatic connections Rilsan® PA11 straight pipe (NOT in spirals), to avoid any expansions phenomena, or add excessive volume to the test circuit.
- For the connection to the TEST PORT with the piece under test use the shortest pipe you can to avoid to add a parasite volume at the testing circuit, that would make increase testing time and expansion phenomena or thermal drift.
- Do not use quick couplings on the measure circuit, because they are not indicated for leak circuits.
- For the connection to the piece under test use only tight –fitting ring.

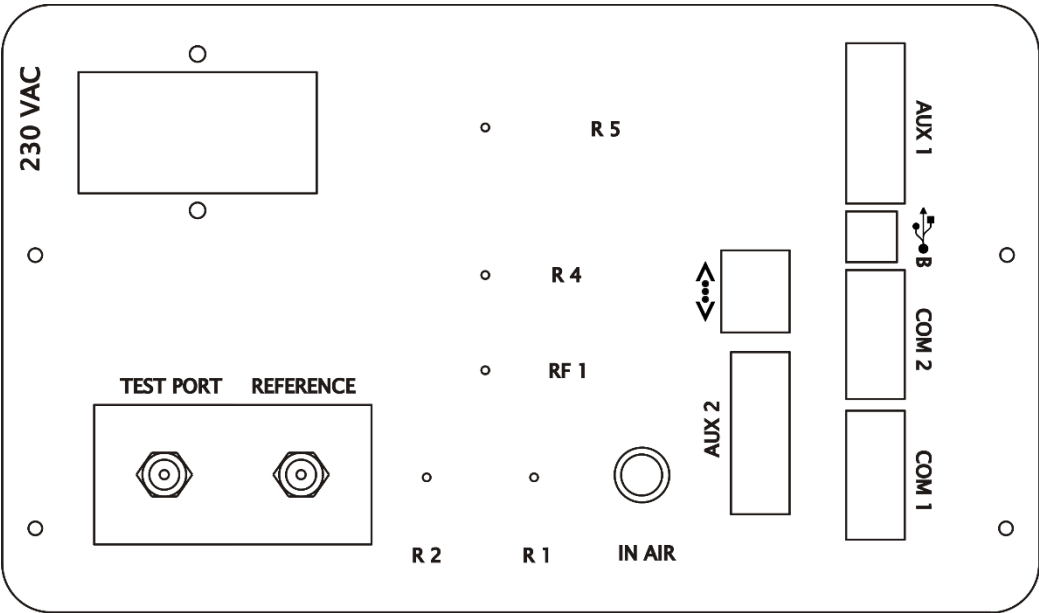


WARNINGS: use gas to a different pressure than the ambient one can cause risks depending by the force that the gas can have. Whatever it is the application, the testing parts, pipes, plugs they can however represent a possible source of risk and of danger for human people and objects. Take care to verify always the hydraulic circuit, to put it under protection with the use of barriers or any other system with this aim. We recommend to train your personnel about the risks caused by the use of gas under pressure, and also about the use of the equipment and how to test the pieces. ForTest guarantees its products safety (if they are used with gas and pressures they have been designed for, and if they work with a qualified staff), but it disclaims any responsibility if they are used in a wrong way or without respecting basic conditions contained in the present manual.

Here below there is the picture with the disposal of pneumatic components present in the back panel of the equipment for the standard version of the equipment (some optional could modify the

components placed on the panel, in this case modifications will be indicated in the chapter concerning the same optional). For electric junctions, refer to the chapter *Back Panel Electric Components*.

Please note: when we act on these components, both electric junctions and mechanical and pneumatic parts, please be careful, in order to avoid to damage them.



IN AIR	Air supply junction: make the connection with filtered compressed air at 25 micron, without oil, without agents or contaminated bodies (chips, dust, etc...). If the full scale pressure of the equipment is under the 6 bar, pneumatic supply has to be 6 bar; if, on the contrary, the equipment has a full scale more than 6 bar, then the pneumatic supply must be 20% higher than this value (for example, if the full scale is of 10 bar, then supply with 12 bar). In the case of depression instrument (model only in depression, not a positive / negative), plug in the vacuum line connection. We recommend a two-stage filter of 25 micron and 1/100 micron before the pneumatic supply of the equipment.
RF1	Flow regulator 1: it allows to set the filling flow.
TEST PORT	Junction connection of the piece under test: connect the piece to be tested, taking care to use a rigid pipe Rilsan® with a length as little as possible. In case of depression equipment, connect filter accessory to avoid that impurity on piece or in the closing system can lay down inside internal valves of equipment and so compromise its air tightness.
REFERENCE	Junction connection of the master reference part used in DIFFERENTIAL and DUAL ABSOLUTE mode, or connection to the second piece to be tested in ZERO-CENTER test mode.

Electric Components of Back Panel

As you can see in the picture of the chapter *Back Panel*, on the back panel there are some electric connectors, identified by initials placed nearby. Here below we describe its characteristics, its meanings, the connector pinout, etc. Information here below refer to configuration of the standard equipment. In case of customization some signal meanings can become different. Electric connectors of the optional expansion card are not described, and they will be treated in the chapter concerning the same optional.

230VAC

Male connector C16 for electric supply of the equipment, at 230VAC 50/60 Hz. Beside electric connector there is ON/OFF button in order to turn on or off the equipment. Moreover, between socket and button there is a fuse housing 4A 250V retarded. The equipment inside has a lot of protections, net and fuse filters, but it is better to feed it with stable and filtered voltage.

Please note: on request there are some Instruments that can work correctly with electric supply of 110VAC.

COM2

Male connector DB9:

I/O optional signals have the same electrical characteristic of signals on AUX1 connector; they have more or less these characteristics:

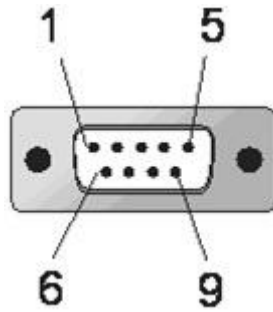
- Out0, Out1: active source exit 24 Vdc (common to GND), able to supply at least 2,5A, internally limited and protected, with firmware management for the diagnostics of exit anomalies. This exit is optional you can enable on a specific inquiry of customer.
- Out Vacuum: active source exit 24 Vdc (common to GND), able to supply at least 2,5A, internally limited and protected, with firmware management for the diagnostics of exit anomalies. This exit is activated during filling phase in case of equipment is testing in vacuum.
- TTY: TTY communication at current loop. It is useful in an industrial environment to provide for interferences and electromagnetic obstacles and to reach long distances of communication (also 1000 m). Moreover, this type of communication allows to realize multiple points network with more equipment on the same ring. So, it is possible, in a very simple way and with easy wiring, to connect in network to check more equipment together also in big productive environment. Internally, TTY signals are optoisolated.

Please note: TTY communication use the same hardware source of RS232 SER1 serial present on COM1 connector.

-OutAn: Analog output. This signal is active only if on the instrument the optional “Analog Output” is activated. For further information see the paragraph regarding this optional.

Please note: In case this optional is present, the TTY communication will not be available.

The following picture shows the pinout of the connector.



COM2	
Pin	Description
1	TTY_T+/ OutAn
2	TTY_R+
3	TTY_R-
4	TTY_T-
5	GND
6	EV FILL
7	OUT1
8	Out Vacuum
9	GND

USB B

USB connector of type B female: it is possible to connect this connector with a classical USB cable of type B, and to communicate with a device by USB protocol. For any communication modes, any communication speed, any communication protocol, please refer to the chapter concerning the serial protocol and to the parameters of Setup Menu. The pinout of the connector is the standard one. Inside the signals are optisolated.

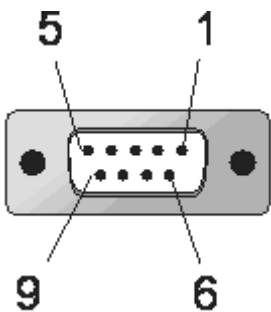
Please note: USB B and COM1 use the same hardware resource, they have only a different connector to make connection. Then it is not possible to use both of them at the same time. In particular, the used source is SER0.

COM1

Female DB9 connector: on this connector there are two modes of communication, that is a classical RS232, RS485 interface and CAN BUS optional interface.

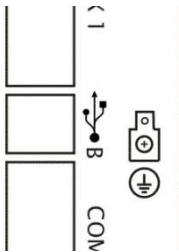
Please note: TTY, RS232 SER0, RS485 and CAN BUS (on COM2 connector) use the same hardware resource, they are only a different way in which communication can be used. Then it is not possible to use both of them at the same time.

For any possible way of communications, any speed of communication, and protocol of communication, please refer to the chapter concerning serial protocol and to the parameters of Setup Menu. The following picture shows the pinout of the connector



COM1	
Pin	Description
1	CANH (OPT)
2	TX RS232 SER1
3	RX RS232 SER1
4	CANL (OPT)
5	GND
6	TX RS232 SER0
7	RX RS232 SER0
8	RS485 B\
9	RS485 A

GROUNDING SYSTEM



Connection for the grounding system (available only for equipment with power supply 24VDC)

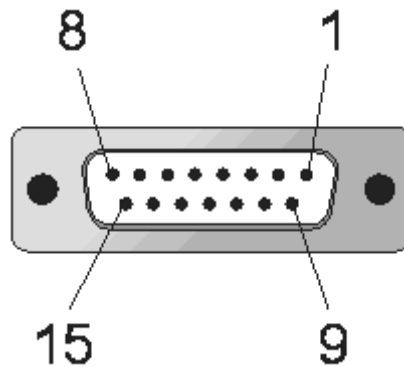
AUX1

Female DB15 connector for connection to PLC or external benches.

Signals that are present on the connector can be grouped into four classes:

- GND: electric mass of signals, used as a common of the outputs.
- 24Vdc: electric supply of signals, used as a common of the input. This common is protected inside with PTC (resettable fuse), able to supply max 750mA.
- Out: 24Vdc active outputs source (common to GND), it is able to supply max 2,5A, inside it is limited and protected, with a firmware management for diagnosis of output anomalies.
- In: 24Vdc active inputs (common at 24Vdc), optisolated, it is able to absorb max 50mA.

Here below there is a picture for the positioning of the number of pin connector and a table with the pinout of the connector marked by initials AUX1.



AUX		
Pin	Description	Signal type
1	GND	GND output
2	GND	GND output
3	GND	GND output
4	START 4	Input
5	TEST	Output
6	REJECT	Output
7	GOOD	Output
8	START 1/BI MANUAL	Input
9	START 2	Input
10	START 3	Input
11	REJECT REFERENCE	Output
12	START	Input
13	ABORT	Input
14	E.V. DISCHARGE	Output
15	24 Vdc	+ 24 Vdc

TEST signals identify the equipment state, distinguishing between test in progress and test not in progress. TEST signal activation depends on the relative setup parameter, “**End of cycle signal:**” and also on the parameter “**Label signal duration:**”: see the chapter *Setup Menu* for explanations.

REJECT signal identifies the waste result of the last test made by the equipment after it has been started. REJECT signal activation depends on parameter setup “**End of cycle signal:**”, on the parameter “**Bad management:**” and also on the parameter “**Label signal duration:**”: see the chapter *Setup Menu* for its explanation.

GOOD signal identifies the good result of last test made by the equipment after it has been started. GOOD signal starting depends on the parameter setup “**End of cycle signal:**” and also on the parameter “**Label signal duration:**”: see the chapter *Setup Menu* for its explanation.

The REJECT REFERENCE signal identifies the reference side reject result (in differential and zero center mode) of the last test performed by the instrument after its power on. The activation of the REJECT REFERENCE signal depends on how the Setup parameter "End of test signal:" has been set, on the parameter "Rejection management:" and also on the parameter "Duration of label signal:": please refer to the Setup Menu chapter for an explanation.

ABORT signal is generally get abort to a test in progress, but it can have different functions depending on the setup parameter “**End of cycle signal:**”: see the chapter *Setup Menu* for its explanation.

Signals START, START1, START2, START3, START4 (START4 is not present on the M6xxx equipment) allow to make a test starting. See the relative parameter “**Kind of start:**” described in the Setup Menu in order to understand different combinations of starting types. To know when the equipment is sensible to a new start, see the parameter description “**End of cycle signal:**”.

Here it is how the signals are combined according to the starting type selected:

- “**Bimanual**”: in this case the equipment is sensible only to signals START and START1; this mode is selectable only if the relative option is active.
- “**Panel excluded**”: in this case the equipment is sensible only to START signal.
- “**From Panel**”: in this case the equipment is sensible only to START signal.
- “**4 Start**”: in this case the equipment is sensible to all the five signals: START1 makes the program 1 starting, START2 makes the program 2 starting, START3 makes the program 3 starting, START4 makes the program 4 starting
- “**BCD+STROBE**”: in this case the equipment is sensible to all the five signals: see here below for BCD weight of to the signals.

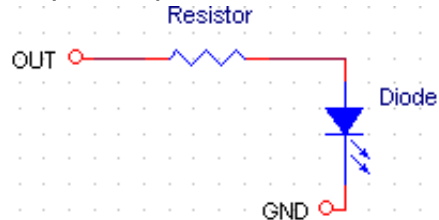
- **“BCD+200ms”**: in this case the equipment is sensible to all the five signals: see here below for BCD weight of to the signals.

START8 (only with I/O esp.)	START7 (only with I/O esp.)	START6 (only with I/O esp.)	START5 (only with I/O esp.)	START4	START3	START2	START1	Matching program number
0	0	0	0	0	0	0	0	----
0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	0	2
0	0	0	0	0	0	1	1	3
0	0	0	0	0	1	0	0	4
0	0	0	0	1	0	0	0	8
0	0	0	1	0	0	0	0	16
0	0	1	0	0	0	0	0	32
0	1	0	0	0	0	0	0	64
1	0	0	0	0	0	0	0	128

Please note: In case of setting “expansion I/O 24 VDC” optional are present 4Auxiliar Start, as shown in table above.

Examples of Electrical Connections

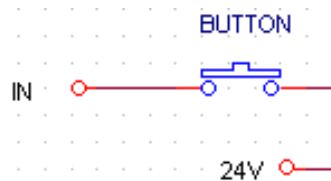
In this chapter we describe some examples of electrical schemes for the signal's connection. For more complex circuits or for PLC interfacing, we highly recommend the use of some devices that decouples the two parts (i.e. some simple relè)



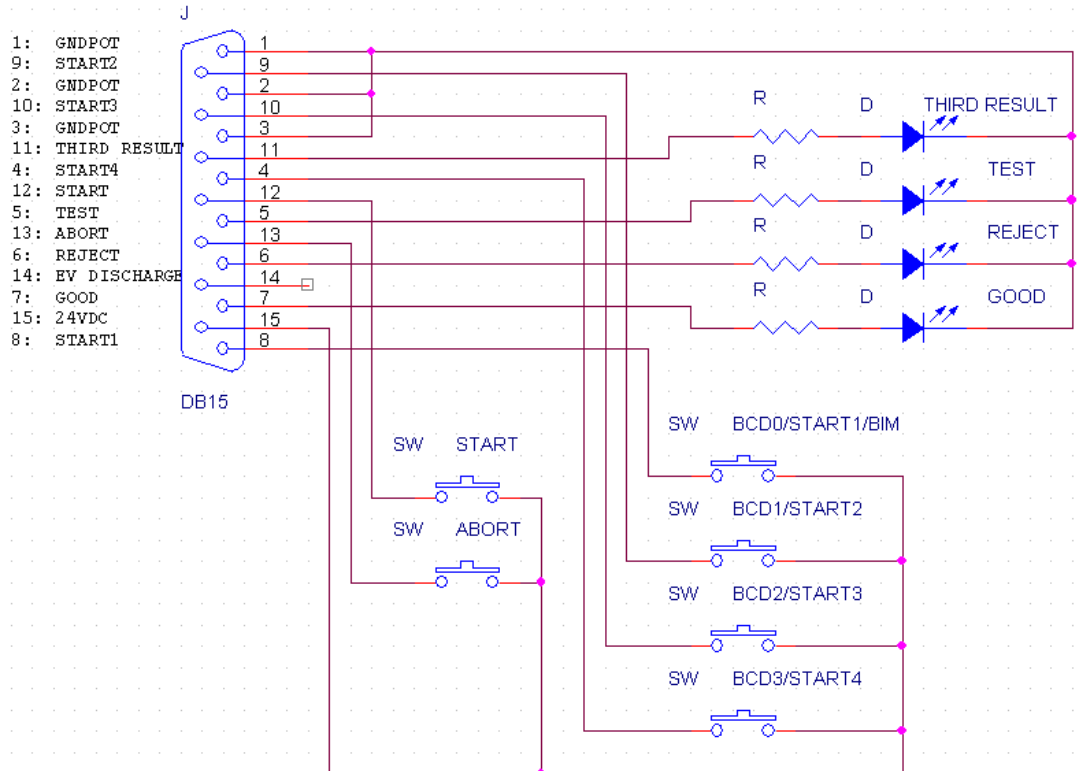
The picture shows a basic scheme for the connection of a led with an outcome output (OUT).

$$R = (24V - V_f) / I_f$$

Where R is the resistance value, 24V is the power supplied by the connector, V_f is the direct voltage of diode led, I_f the current of diode led.



The picture shows the basic scheme for the connection of a button to the input of an equipment (IN).



The picture shows a possible complete bench but almost simple, to exploit all the I/O supplied by the connector AUX1 of the instruments T9xxx, T8xxx.

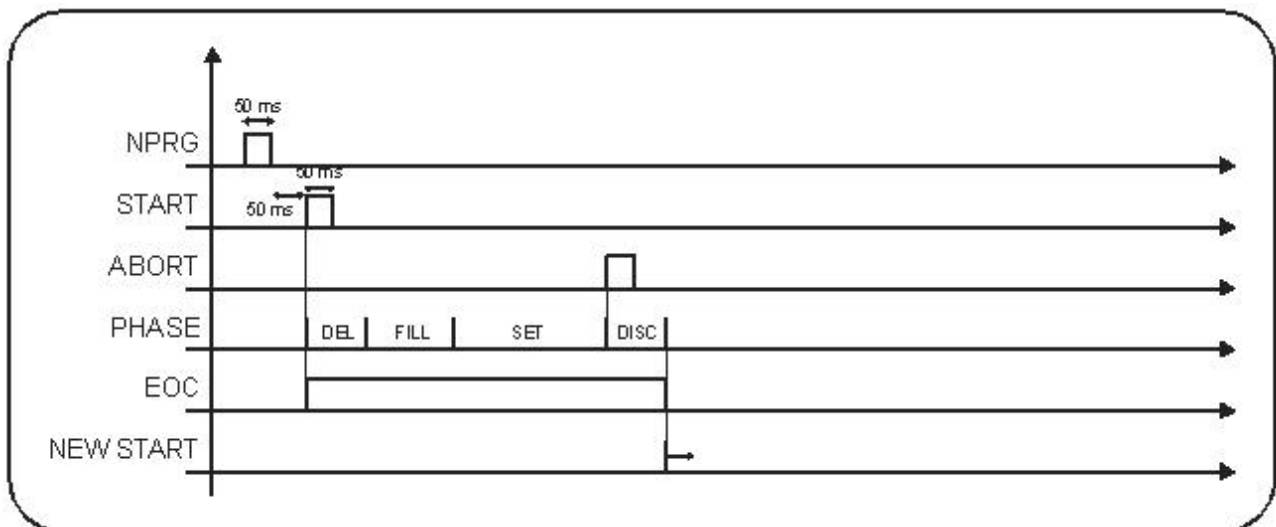
Timing Diagrams

In this chapter there are some pictures that represent timing trends of signals that are on the AUX1 connector. These diagrams and their signal function change according to their parameter setup **“Label:”/ “Label signal duration:”**, **“EOC:”/ “End of cycle signal:”**, **“Bad:” / “Bad management:”**, **“Abort:”/ “Abort management:”**. In this chapter we realize only timing diagrams for a better understanding of the description already given in the explanation of the same parameters in the chapter *Setup Menu*.

Generally, in order to recognize input signal by the equipment, it is necessary that these ones remain activated for at least 50ms. In case you want to send two different input signals (for example program of setup in **“BCD+S” / “BCD+STROBE”** mode and its consequent start), they have to be distant temporarily at least 100ms between their rising edges.

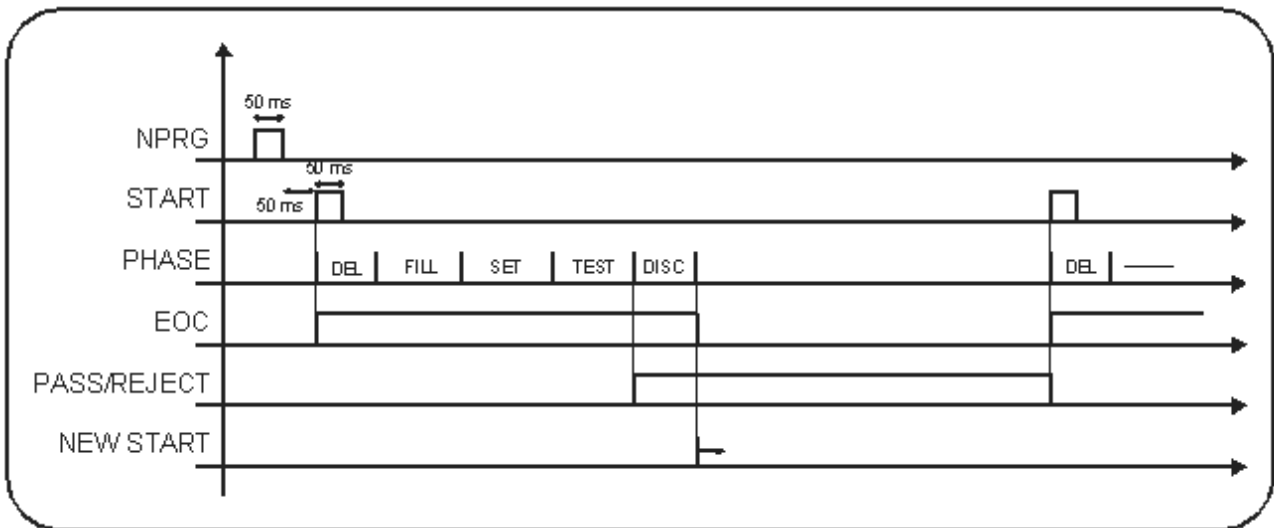
Standard Configuration in case of abort:

“EOC:” / “End of cycle signal:”	“Std” / “Standard”
“Abort:” / “Abort management:”	“Std” / “Standard”



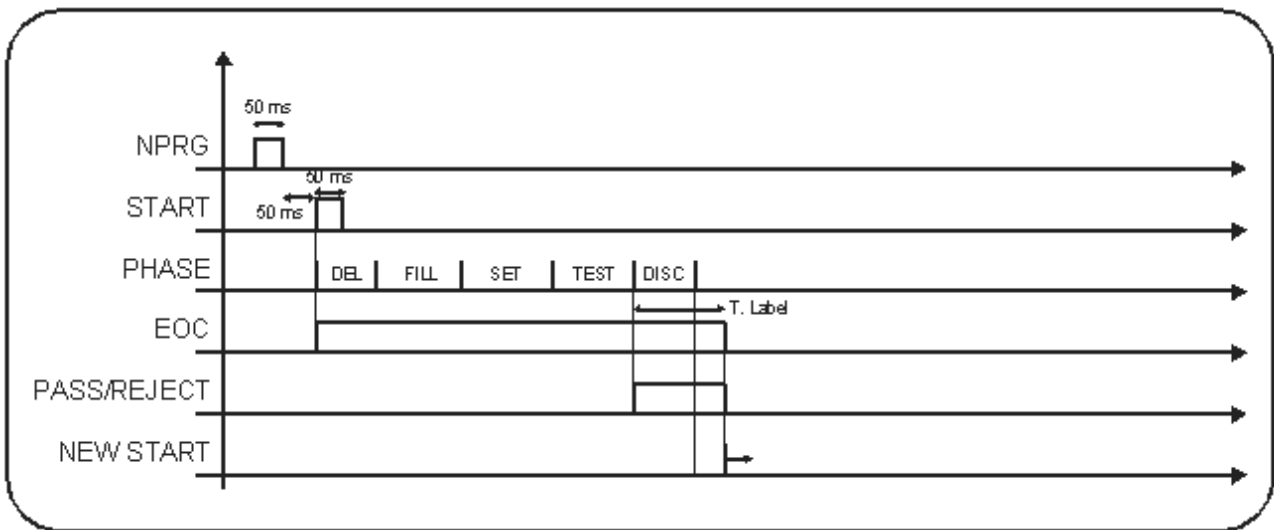
Standard Configuration in case of waste/good with null time:

“EOC:” / “End of cycle signal:”	“Std” / “Standard”
“Bad:” / “Bad management:”	“Std” / “Standard”
“Label:” / “Label signal duration:”	“0.0”



Standard Configuration in case of waste/good with time different from zero:

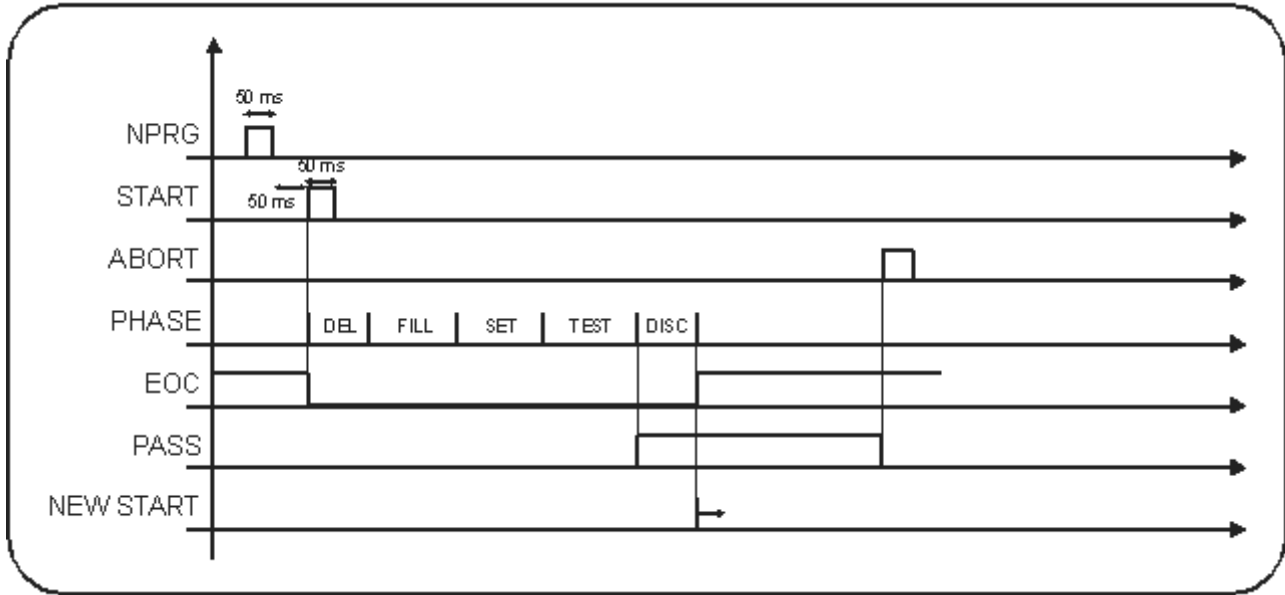
“EOC:” / “End of cycle signal:”	“Std” / “Standard”
“Bad:” / “Bad management:”	“Std” / “Standard”
“Label:” / “Label signal duration:”	“2.0”



To note that in the example the discharge is quickly than **“Label:” / “Label signal duration:”**: in the opposite case, the signal **“EOC:” / “End of cycle signal:”** would have been disabled at the end of the discharge (and at the same time a new start would have been possible), while **“Bad:” / “Bad management:”** are disabled in both the two case at the end of **“Label:” / “Label signal duration:”**.

Non-Standard configuration with end cycle ready and case of good result:

“EOC:” / “End of cycle signal:”	“Ready” / “Ready”
“Bad:” / “Bad management:”	Different from “Std” / “Standard”

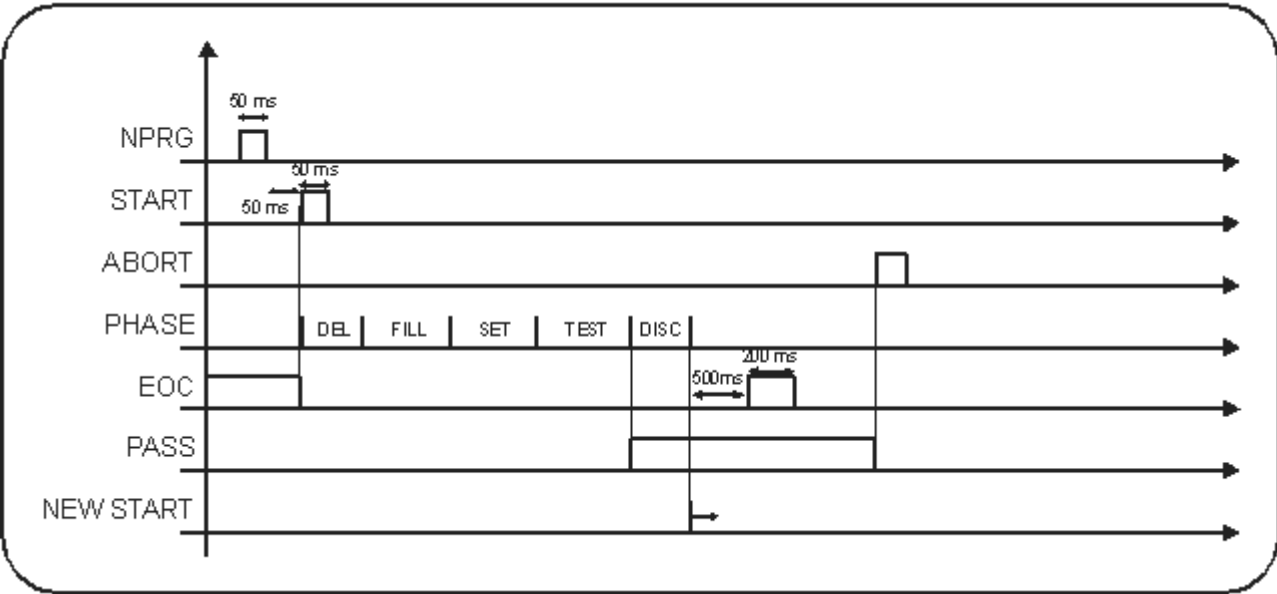


Whatever it is the parameter set **“Bad:” / “Bad management:”**, in case of good as result the behaviour is that one described (it coincides with a waste in case of **“Bad:” / “Bad management:”** has the value **“Auto:” / “All Automatic:”**).

In the case the **“EOC:” / “End of cycle signal:”** has the value **“Busy” / “Busy”** it is totally the same, unless the signal **“EOC:” / “End of cycle signal:”** has a dual working (i.e. it is low out of the test and high during the test).

Not Standard configuration with impulsive end of cycle and case of good result:

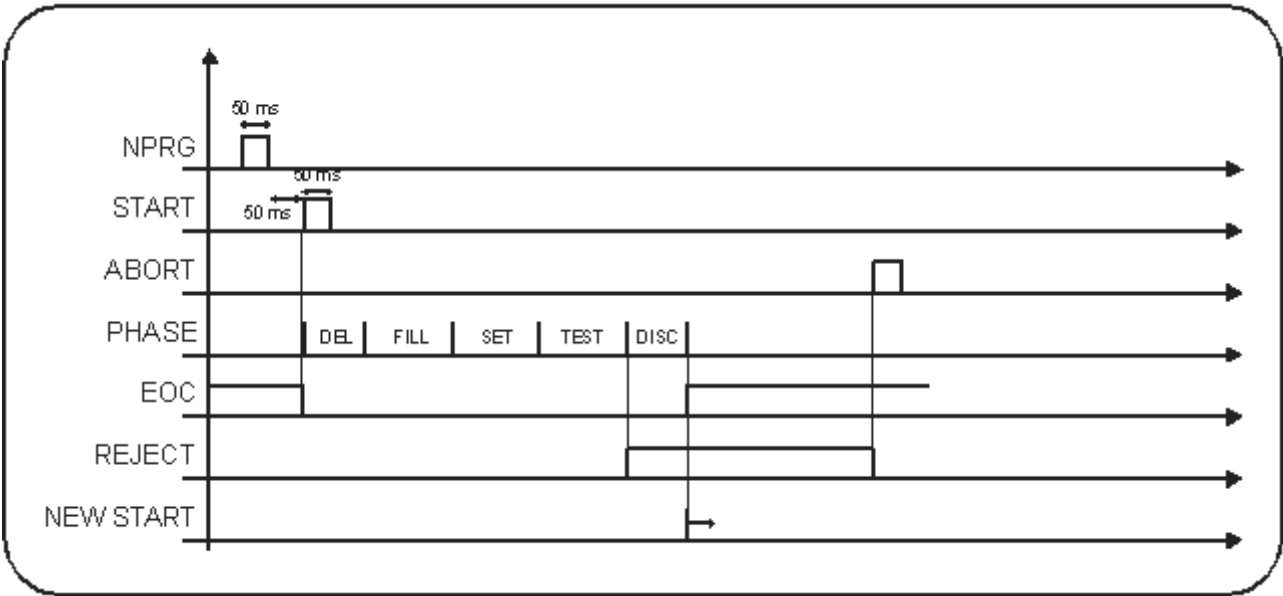
“EOC:” / “End of cycle signal:”	“Pulse” / “Pulse”
“Bad:” / “Bad management:”	Different from “Std” / “Standard”



Whatever it is parameter set **“Bad:” / “Bad management:”**, in case of good result the behaviour is that one described (it coincides with a waste in case of **“Bad:” / “Bad management:”** has the value **“Auto:” / “All Automatic:”**).

All automatic configuration with end of cycle ready and case of waste result:

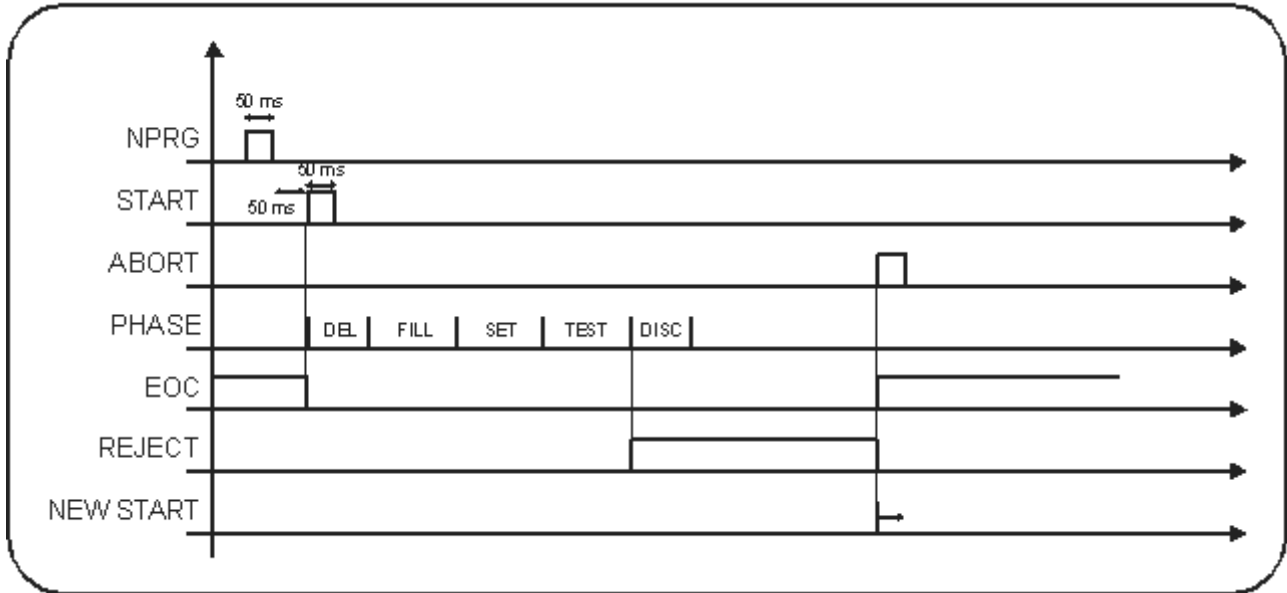
“EOC:” / “End of cycle signal:”	“Ready” / “Ready”
“Bad:” / “Bad management:”	“Auto” / “All Automatic”



In this case the abort signal serves to disarm waste output, that obviously it would be disarmed also by a new start. In the case **“EOC:” / “End of cycle signal:”** has the value **“Busy” / “Busy”** is totally the same, unless that the signal **“EOC:” / “End of cycle signal:”** has a dual working (i.e. it is low out of the test and high during the test).

Configuration validation with end of cycle ready and case of waste as result:

"EOC:" / "End of cycle signal:"	"Ready" / "Ready"
"Bad:" / "Bad management:"	"Val" / "Validation"

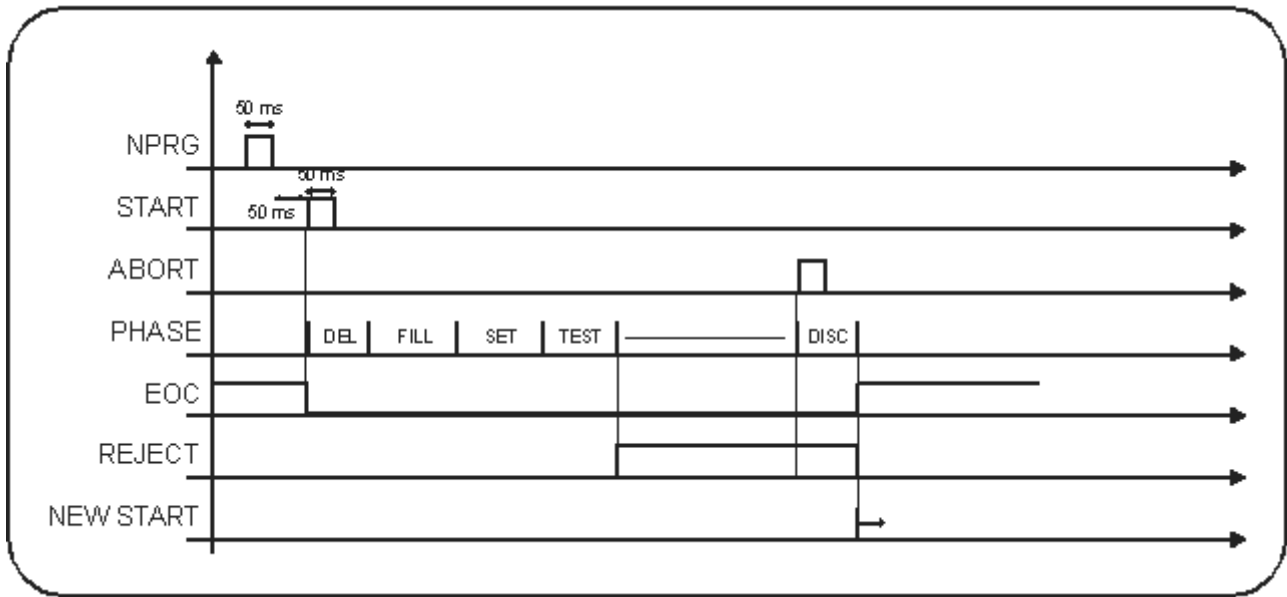


In this case the abort signal serves to validate the test: in fact, it decrees the end of the test, the deactivation of waste signal, and the possibility of giving a new start. Since the equipment does not receive the abort signal it remains fixed with waste result without allowing to make new tests.

In case **"EOC:" / "End of cycle signal:"** has the value **"Busy" / "Busy"** it is totally the same, unless that the signal **"EOC:" / "End of cycle signal:"** has a dual working (i.e. it is low out of the test and high during the test).

Configuration leak detection with end of cycle ready and case of waste as result:

"EOC:" / "End of cycle signal:"	"Ready" / "Ready"
"Bad:" / "Bad management:"	"Full" / "Remain full"

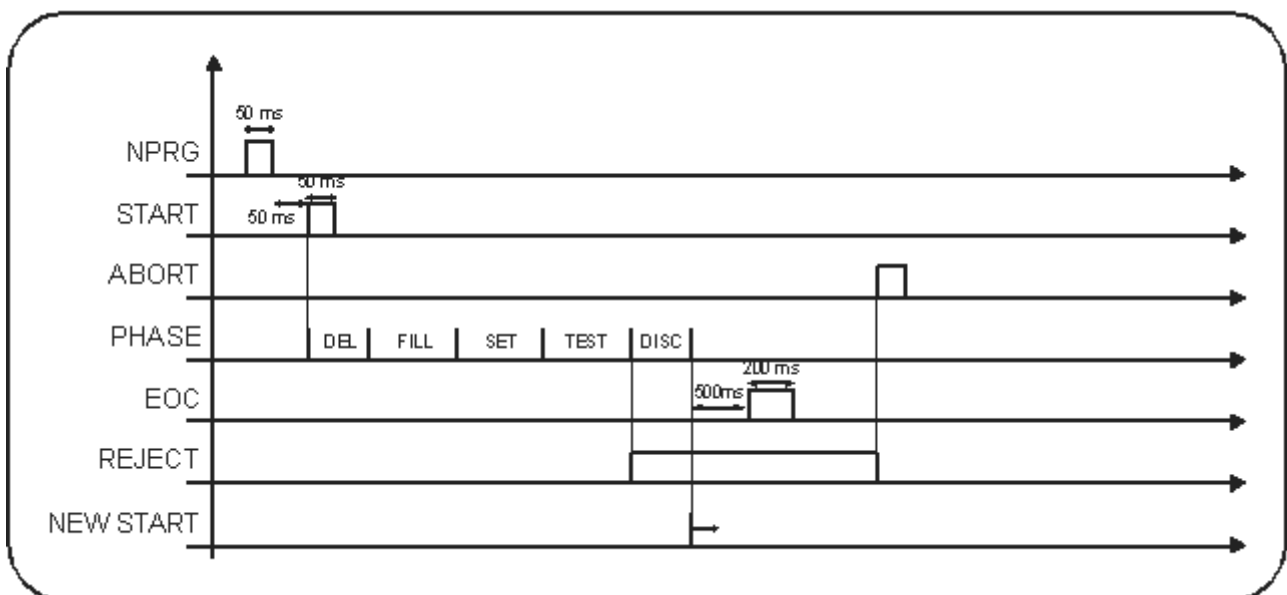


In this case abort signal serves to start the discharge and then to end the test: in fact at the end of the test phase, the discharge does not start in an automatic way leaving under pressure the testing piece (this can be particularly useful in case you want to detect, by soap methods or other ones where is the leak). When the equipment has received the abort signal it starts the discharge phase and the following activations and deactivations of signals.

In case **"EOC:" / "End of cycle signal:"** has a value **"Busy" / "Busy"** it is totally the same, unless that the signal **"EOC:" / "End of cycle signal:"** has a dual working (i.e. it is low out of the test and high during the test).

All automatic configurations with impulsive end of cycle and case of waste:

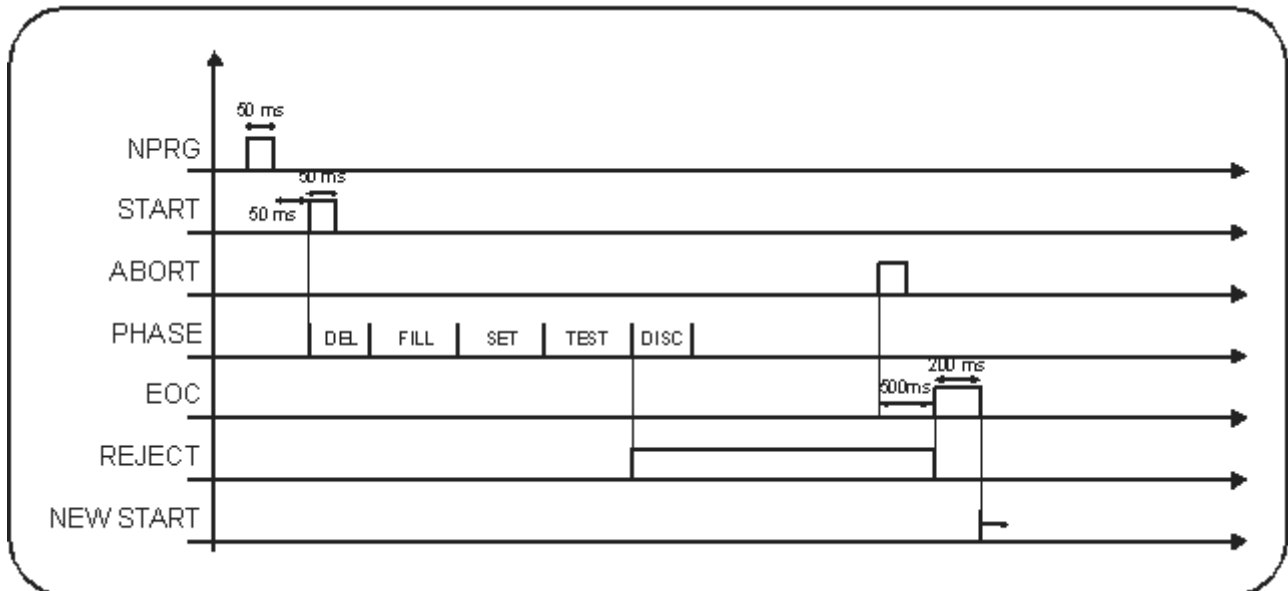
"EOC:" / "End of cycle signal:"	"Pulse" / "Pulse"
"Bad:" / "Bad management:"	"Auto" / "All Automatic"



In this case the abort signal serves to disarm the waste output, that obviously it would be disarmed also by a new start.

Configuration validation with impulsive end of cycle and case of waste:

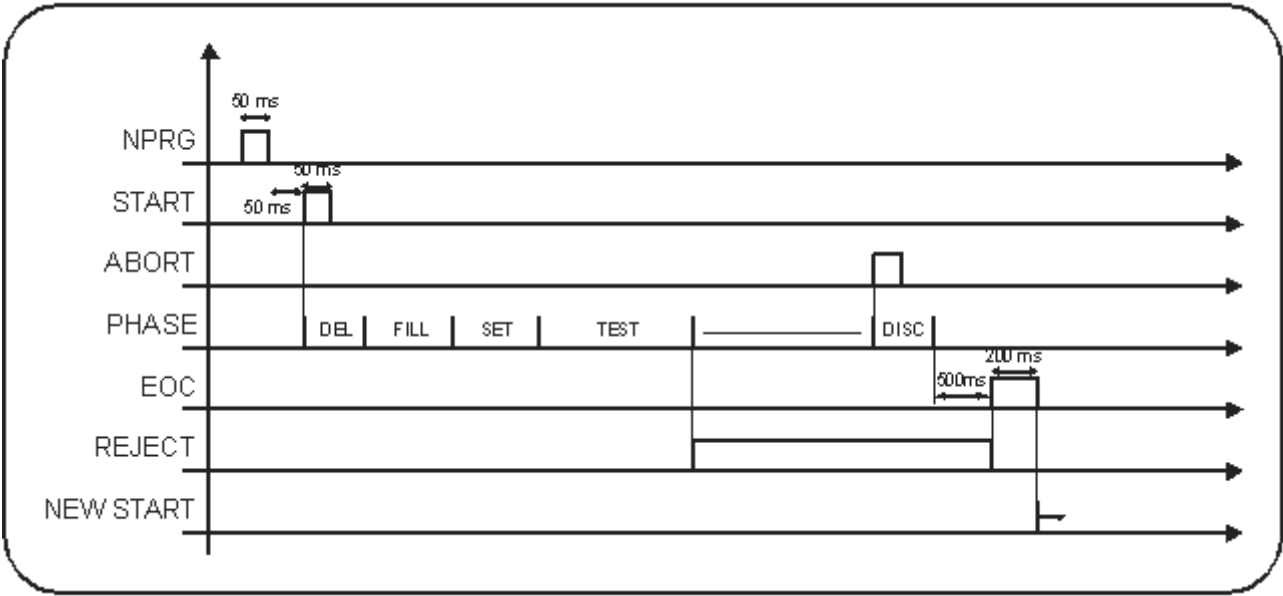
"EOC:" / "End of cycle signal:"	"Pulse" / "Pulse"
"Bad:" / "Bad management:"	"Val" / "Validation"



In this case the abort signal serves to validate the test; in fact, it decrees the end of the test, the deactivation of waste signal, and the possibility of giving a new start. Since the equipment does not receive the abort signal it remains fixed with waste as result without allowing to make any other tests.

Configuration of leak detection with impulsive end of cycle and case of waste:

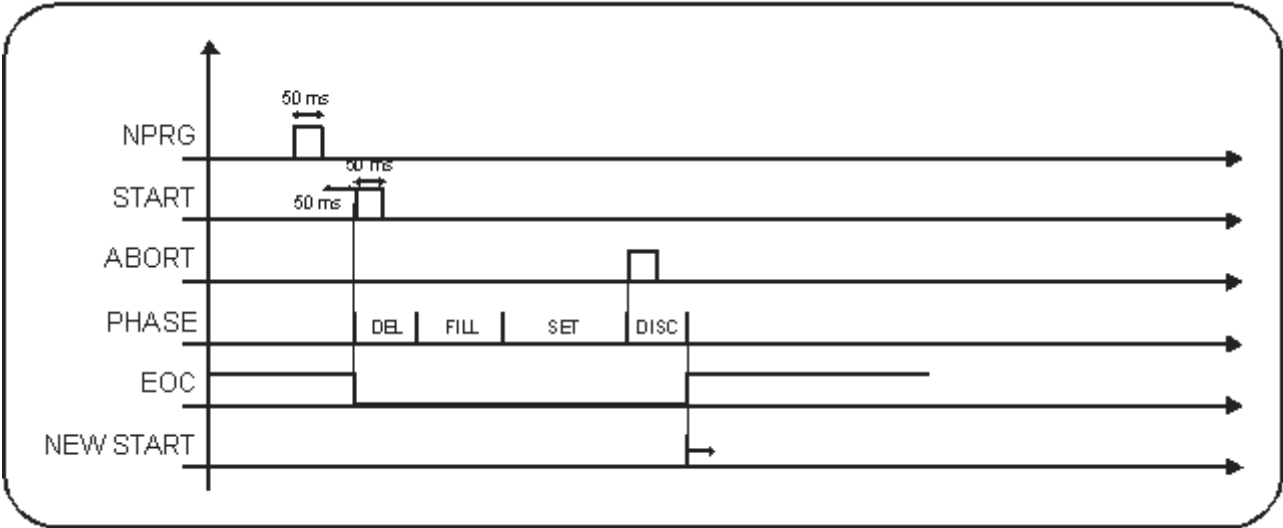
"EOC:" / "End of cycle signal:"	"Pulse" / "Pulse"
"Bad:" / "Bad management:"	"Full" / "Remain Full"



In this case the signal of abort serves to start the discharge and then to end the test phase: in fact at the end of the testing phase the discharge does not start automatically leaving under pressure the testing piece (this can be particularly useful in case you want to detect, by soap methods or other ones where is the leak). When the equipment has received the abort signal it starts the discharge phase and the following activations and deactivations of signals.

All automatic configuration with end of cycle ready and case of abort:

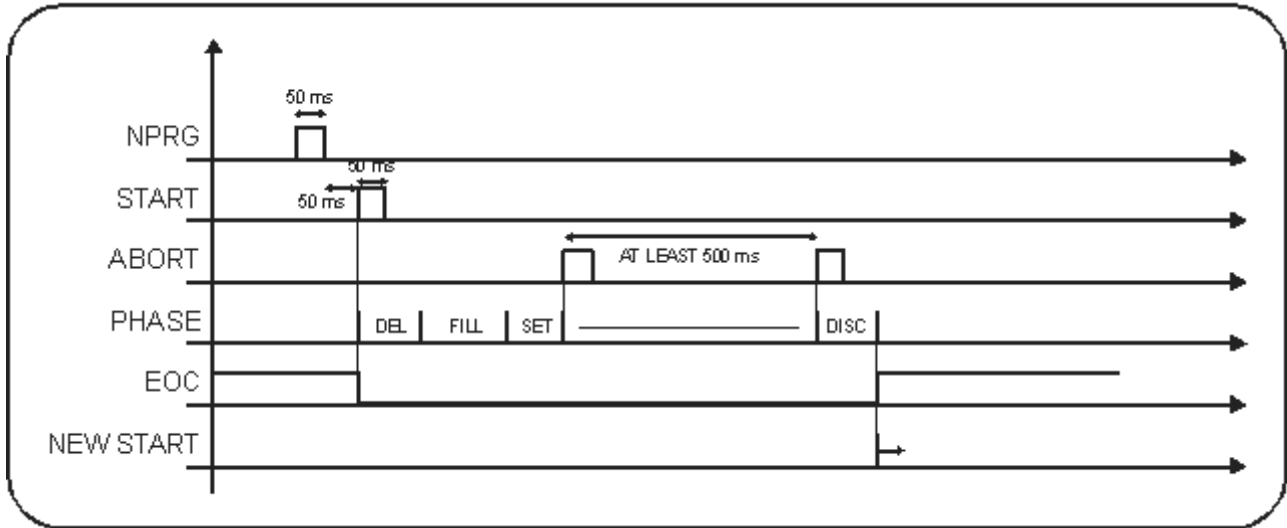
"EOC:" / "End of cycle signal:"	"Ready" / "Pronto"
"Abort:" / "Abort management:"	"Auto" / "All automatic"



In case "EOC:" / "End of cycle signal:" has a value "Busy" / "Busy" it is totally the same, unless that the signal "EOC:" / "End of cycle signal:" has a dual working (i.e. it is low out of the test and high during the test).

Leak detection configuration with end of cycle ready and case of abort:

"EOC:" / "End of cycle signal:"	"Ready" / "Ready"
"Abort:" / "Abort management:"	"Full" / "Remain Full"

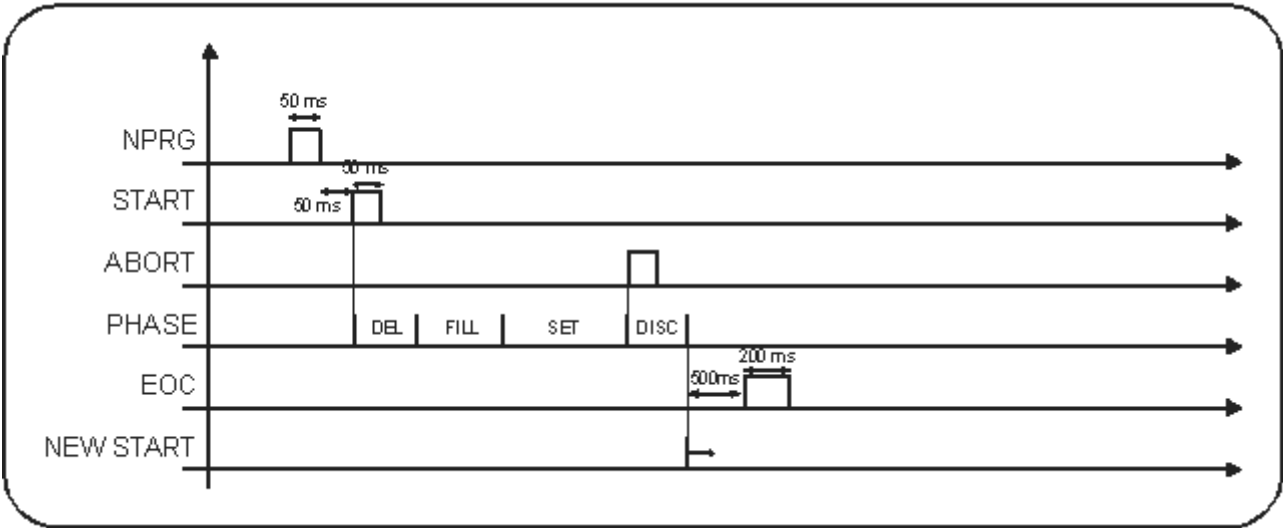


In this case a second signal of abort (that has to be temporarily away from the first one of at least 500ms) serves to start the discharge and then to end the test: in fact after the first abort signal the discharge does not start automatically leaving the testing piece under pressure (this can be particularly useful in case you want to detect, by soap methods or other ones where is the leak). When the equipment has received the second abort signal it starts the discharge phase and the following activations and deactivations of signals.

In case **"EOC:" / "End of cycle signal:"** has a value **"Busy" / "Busy"** it is totally the same, unless that the signal **"EOC:" / "End of cycle signal:"** has a dual working (i.e. it is low out of the test and high during the test).

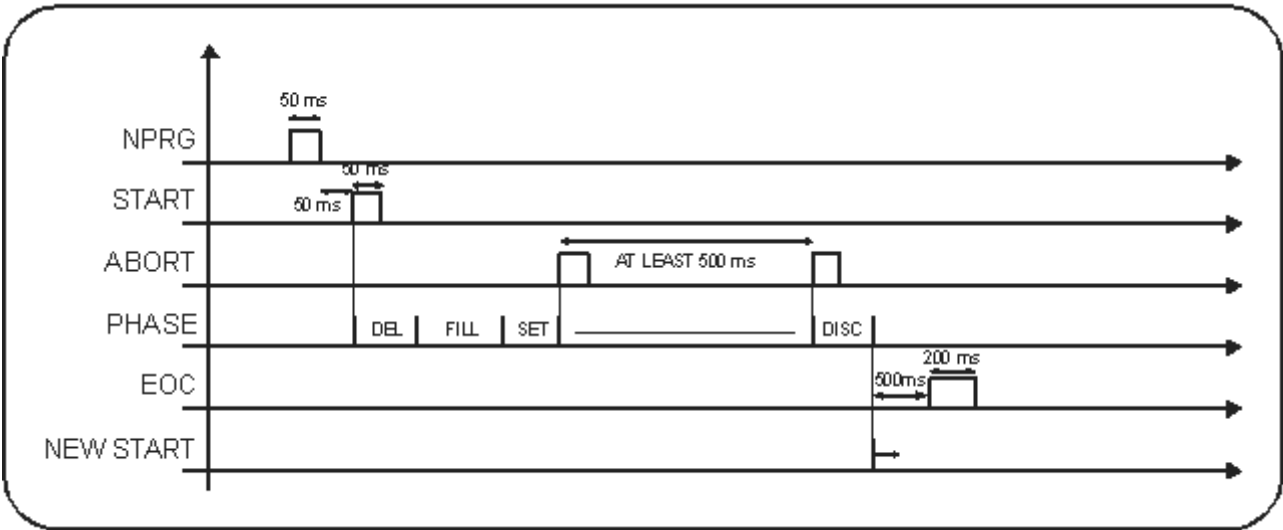
All automatic configuration with impulsive end of cycle and case of abort:

"EOC:" / "End of cycle signal:"	"Pulse" / "Pulse"
"Abort:" / "Abort management:"	"Auto" / "All Automatic"



Leak detection configuration with impulsive end of cycle and abort case:

“EOC:” / “End of cycle signal:”	“Pulse” / “Pulse”
“Abort:” / “Abort management:”	“Full” / “Remain Full”



In this case a second signal of abort (that has to be temporarily away from the first one of at least 500ms) serves to start the discharge phase and then to end the test: in fact after the first abort signal the discharge phase does not start automatically leaving the testing piece under pressure (this can be particularly useful in case you want to detect, by soap methods or other ones where is the leak). When the equipment has received the second abort signal it starts the discharge and the following activations and deactivations of signals.

Regular Maintenance of the Equipment

Regularly the equipment has to be checked in order to avoid malfunctions. The most important checks have to be divided between the daily ones or the periodic ones.

REGULAR CHECK OF LEAK TESTERS: LEAK CONTROL OF THE TESTER

To verify in a correct way the pneumatic leak of an equipment, follow these procedures, regularly every 6 months:

1. Set test parameters, that are different according to the full scale.

Ex.1

Full scale equipment 6000.0 mbar:

Filling pressure: 5500.0 mbar

Setting time: 300.0 s.

Test time: 60.0 s.

Max pressure decay: 1.000 mbar

Ex.2

Full scale equipment 1000.0 mbar:

Filling pressure: 900.0 mbar

Setting time: 300.0 s.

Testing time: 60.0 s.

Max pressure decay: 0.500 mbar

Looking to the two examples mentioned above, it is easy to understand that filling pressure setting change according to the full scale of the equipment, and exactly it must be lower than some hundreds of mbar.

2. Connect to the TEST PORT junction a cap with the smallest volume that is possible (clearly the cap has to be tight).
3. Reduce the pressure turning counter clockwise the pressure regulator placed on the frontal panel (see chapter *Frontal Panel*), in order to avoid excessive pressure bump due to a volume almost at zero.
4. Make the test starting, raising the pressure regulator till arriving to the set pressure.
5. Make more times leak test, removing and hanging up from time to time the pipe under test, controlling that the equipment give a good result.

N.B: For flow test instruments or digital calibrators, you have not to test the leak, because, measuring a flow, they do not need this procedure.

REGULAR CONTROL FOR ALL THE EQUIPMENTS: CONTROL OF THE CLEANING OF THE EQUIPMENT

Control regularly (every 3 months) equipment cleaning:

- Plexiglas (or aluminium front case) must be cleaned regularly with a cloth moistened with warm water.
- Control any aluminium button are not seized for presence of dust, metal debris, etc...
- Clean the box in anodized aluminium regularly with a cloth moistened with warm water to prevent the entry of contaminants inside the equipment.
- If present, clean the back-aluminium panel to avoid the entry of contaminants inside the equipment.

REGULAR CONTROL FOR ALL THE EQUIPMENTS: CLAMPING SCREWS

Regularly verify (every 6-12 months) the correct clamping screws of the equipment:

- Control the 4 frontal screws securing plexiglass (or aluminium frontal), if loosen, softly tighten them.
- Control 4 fixing screws of back panel (if present). If loosen, close them softly.

DAILY CONTROL OF THE EQUIPMENT: VERIFY CONTROL SYSTEM

As recommended by the standard ISO it is better to have two referring samples corresponding one to the good and the other to the waste.

Really, analysing better ISO procedure, and considering that the leak measured has to be understand as a limit of tolerance T not acceptable (obviously as a term of leak this tolerance cannot be symmetrical but only as loss), the ideal procedure is to place a sure good piece and to place both the frontal panel Staubli option (see chapter *Frontal Panel*) both the calibrated leak M0710 at the loss level (T) of the application in object and then to superimpose at the good piece to obtain a sure value of waste.

Therefore, before every cycle of control you have to verify that the equipment detects correctly the two limits, using two referring samples and repeating different tests, alternating the good sample to the calibrated leak. Make the two-sample rest for some minute between a test and the other one, to avoid phenomena of temperature, expansion, etc...

Problem Solving

Problem	Solution
The instrument does not start.	Verify that power cable is connected. In case of instrument with rechargeable battery, recharge it with the recharger.
The instrument does not start.	Verify that the fuse in the net filter is complete. (Fuses 4 A – 250V delayed in glass 5X20 mm.) Only for equipment at 230VAC.
The instrument does not start and both the fuses and the power cable are ok. The battery is charged.	Send the instrument back to an assistance point.
The graphic display starts but very dim.	Verify the parameter that regulates its brightness in the Setup Menu.
The graphic display is on has a good brightness, but messages are not readable.	Verify the parameter that regulates contrast in the Setup Menu (if you cannot see anything on the front side, it could be sufficient to see something, to look at the display with a better view corner).
The display does not start.	Send the instrument back to an assistance point.
The instrument remains blocked at its starting making a persistent acoustic signal.	Send the instrument back to an assistance point.
The instrument has not a perfect tight.	Check the entireness of the pipe connected to the Test-Port of the instrument and any other plugs of the piece under test.
The instrument presents not a perfect tight even after the control of the pipes and of the plugs.	Send the instrument back to an assistance point.
The instrument does not read the pressure injected in a right way.	Go on with the calibration of the internal transducer of pressure (see the chapter <i>Calibration Menu</i>).
The test does not start pushing the button [+].	Verify that the relative setup parameter has been correctly set.

The instrument does not answer to the instructions sent by the serial protocol.	Verify that the serial address of the instrument (you can regulate it in the Setup Menu) is congruent with that one sent in the instructions.
The instrument does not answer to the instructions sent by the serial protocol.	Check to be connected with the right port COM or USB port.
The instrument does not answer to the instructions sent by the serial protocol.	Check you have set correctly the communication baud rate (in the Setup Menu).
In the filling phase the instrument does not succeed in pressurize the piece.	Verify you are really feeding with compressed air the IN-AIR junction.
In the filling phase the instrument does not succeed in pressurize the piece.	Verify the pressure regulator.

Utilities

Pressure unit table with conversion

Unità di misura Measuring unit	Simbolo Symbol	Pa	bar	at	mm Hg	kgf/m ²	psi	lbf/ft ²	in w.	in Hg	ft w.
pascal	Pa	1	10 ⁻⁵	1,0197x10 ⁻⁵	0,0075	0,10197	0,145x10 ⁻³	0,02088	0,00401	0,295x10 ⁻³	0,335x10 ⁻³
bar	bar	10 ⁵	1	1,0197	750,07	10197	14,505	2088	401,46	29,530	33,456
atmosfera =kgf/cm ² atmosphere	at	98070	0,9807	1	735,56	10000	14,223	2048,16	393,71	28,960	32,808
millimetri di Hg millimetre of mercury	mm Hg	133,32	1,3332x10 ⁻³	1,3595x10 ⁻³	1	13,595	0,0193	1,392	0,5353	0,0394	0,0446
chilogrammi per m ² kilogramme per sq. m	kgf/m ²	9,807	9,807x10 ⁻⁵	10 ⁻⁴	0,0735	1	0,00142	0,205	0,0394	0,0029	0,0033
libbre per pol- lice quadro pounds per sq. inch	psi	6894,14	0,06894	0,0703	51,719	703,07	1	144	27,683	2,0362	2,3069
libbre per piede quadro pounds per sq. foot	lbf/ft ²	47,876	4,7876x10 ⁻⁴	4,8824x10 ⁻⁴	0,7183	4,8824	0,00694	1	0,1922	0,01414	0,01602
pollici di c.a. inches of water	in w.	249,09	0,00249	0,00254	1,868	25,4	0,03614	5,203	1	0,07355	0,0833
pollici di mercurio inches of mercury	in Hg	3386,36	0,03386	0,03453	25,4	345,34	0,4912	70,731	13,595	1	1,1329
piedi di c.a. feet of water	ft w.	2989	0,02989	0,03048	22,42	304,8	0,4334	62,43	12	0,8827	1

Conversion table of volumetric measures (y= nx)

X \ Y	Pa · m ³ /s	mbar · l/s oppure bar · cm ³ /s	Pa · l/s	torr · l/s	lusec (micron l s ⁻¹)	μ · ft ³ /s (micron ft ³ s ⁻¹)	atm · ft ³ /min oppure ft ³ (NTP)/min	std cm ³ /s	molecule/s	mol/s
Pa · m ³ /s	1	10	1 × 10 ³	7,5	7,5 × 10 ³	2,65 × 10 ²	2,097 × 10 ⁻²	9,87	2,651 × 10 ²⁰	4,403 × 10 ⁻⁴
mbar · l/s oppure bar · cm ³ /s	1 × 10 ⁻¹	1	1 × 10 ²	7,5 × 10 ⁻¹	7,5 × 10 ²	2,65 × 10 ¹	2,097 × 10 ⁻³	9,87 × 10 ⁻¹	2,651 × 10 ¹⁹	4,403 × 10 ⁻⁵
Pa · l/s	1 × 10 ⁻³	1 × 10 ⁻²	1	7,5 × 10 ⁻³	7,5	2,65 × 10 ⁻¹	2,097 × 10 ⁻⁵	9,87 × 10 ⁻³	2,651 × 10 ¹⁷	4,403 × 10 ⁻⁷
torr · l/s	1,33 × 10 ⁻¹	1,333	1,33 × 10 ²	1	1 × 10 ³	3,53 × 10 ¹	2,795 × 10 ⁻³	1,316	3,535 × 10 ¹⁹	5,87 × 10 ⁻⁵
lusec	1,33 × 10 ⁻⁴	1,33 × 10 ⁻³	1,33 × 10 ⁻¹	1 × 10 ⁻³	1	3,53 × 10 ⁻²	2,795 × 10 ⁻⁶	1,32 × 10 ⁻³	3,535 × 10 ¹⁶	5,87 × 10 ⁻⁸
μ · ft ³ /s	3,76 × 10 ⁻³	3,76 × 10 ⁻²	3,76	2,83 × 10 ⁻²	2,83 × 10 ¹	1	7,92 × 10 ⁻⁵	3,71 × 10 ⁻²	1,001 × 10 ¹⁸	1,662 × 10 ⁻⁶
atm · ft ³ /min oppure ft ³ (NTP)/min	4,78 × 10 ¹	4,78 × 10 ²	4,78 × 10 ⁴	3,58 × 10 ²	3,58 × 10 ⁵	1,267 × 10 ⁴	1	4,72 × 10 ²	1,268 × 10 ²²	2,105 × 10 ⁻²
std cm ³ /s	1,013 × 10 ⁻¹	1,013	1,013 × 10 ²	7,6 × 10 ⁻¹	7,6 × 10 ²	2,67 × 10 ¹	2,12 × 10 ³	1	2,687 × 10 ¹⁹	4,461 × 10 ⁻⁵
molecule/s	3,77 × 10 ²¹	3,77 × 10 ²⁰	3,77 × 10 ¹⁸	2,83 × 10 ²⁰	2,83 × 10 ¹⁷	9,99 × 10 ⁻¹⁹	7,88 × 10 ⁻²³	3,72 × 10 ²⁰	1	1,66 × 10 ⁻²⁴
mol/s	2,271 × 10 ³	2,271 × 10 ⁴	2,271 × 10 ⁶	1,703 × 10 ⁴	1,703 × 10 ⁷	6,016 × 10 ⁵	4,749 × 10 ¹	2,24 × 10 ⁴	6,022 × 10 ²³	1

Conversion from mass flow to volumetric one

The following formula can be used in the conversion from pV flow to mass flow for a particular gas:

$$q_M = q_V \times M / RT$$

where,

- q_M is the mass flow expressed in kg / s;
- M is the molar mass, expressed in kg for mass;
- q_V is the passing flow, expressed in Pa m³ / s;
- R is the universal constant gas = 8,314 J K mass;
- T is the temperature expressed in K

Volumes calculation

Pipe volume calculation (cylinder)	Parallelepiped volume calculation	Sphere volume calculation
$r^2 \times \pi \times h$ where r: Internal pipe ray π: Greek pi h: total length of pipe	$l \times w \times h$ where l: length w: wide h: high	$r^3 \times 4/3 \pi$ where r: sphere ray π: Greek pi

Technical Details

- Case dimensions 270x160x300 mm, pressure and regulator connectors excluded
- Tempered glass frontal panel
- 230 Vac Power supply
- 10 led RGB to indicate testing results
- 9 command buttons, 8 capacitive touch and 1 mechanical button
- Two bars of 22 blue led to indicate pressure and measure
- Graphic coloured Display 800x280
- Parameters memory EEPROM 4k plus external memory not volatile 32k, with control algorithms
- Hour date RTC buffer battery (time of information retention higher than 250000 hours)
- Outputs at 24Vdc source type (with gnd common) from 2,5 A, protected against over power over temperature and short circuit.
- Optisolated inputs at 24Vdc 10 mA (with common at 24Vdc)
- 2 RS232 communication ports optisolated towards microprocessor side to communicate with PC
- 1 USB port A type to save data on USB key.
- 1 communication port RS485 optoisolated type
- 1 communication port at current loop 0...20 mA TTY Siemens passive optisolated
- 1 communication port CAN BUS optional type
- 1 Bluetooth 2.0/BLE card

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Repair Request Form

Please copy this page and add to the equipment that has to be separated.

Customer	
Address	
Cap.	
City	
State	
Telephone	
Fax	
e-mail	
Reference	
Model	
Serial Number	
Detected defect:	

Send with the equipment to:

ForTest Italia s.r.l.

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